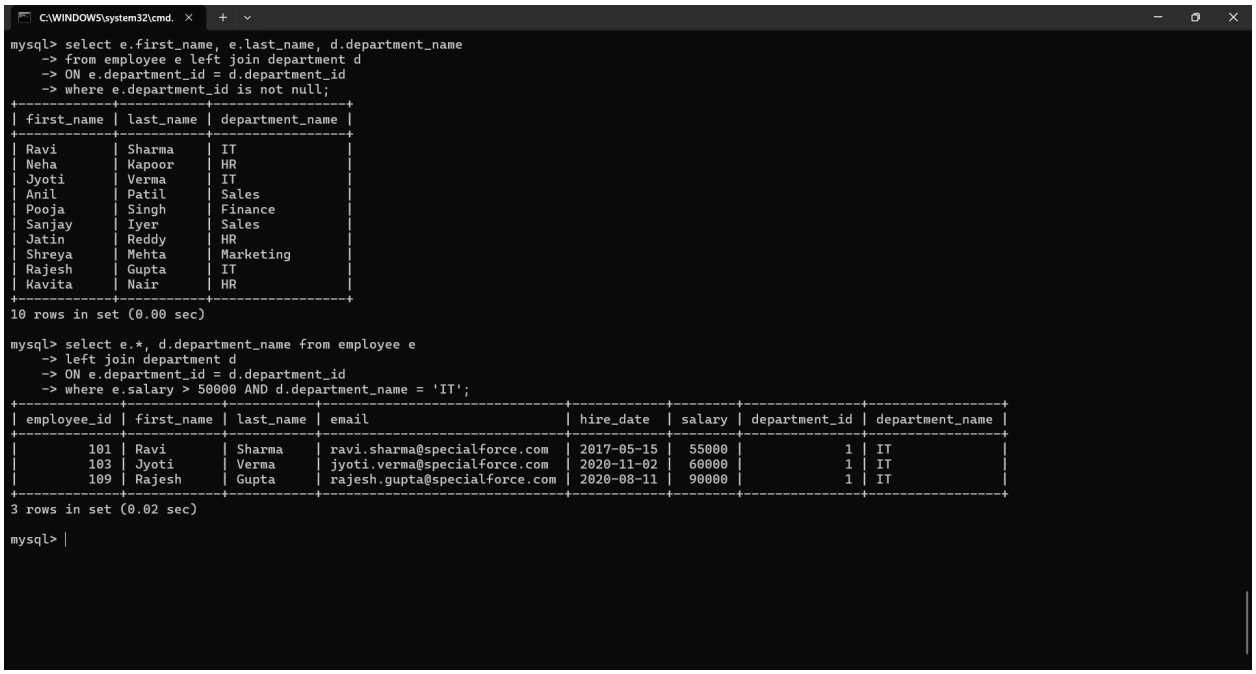
AKSF Assignment 1

Query 1



Query 2



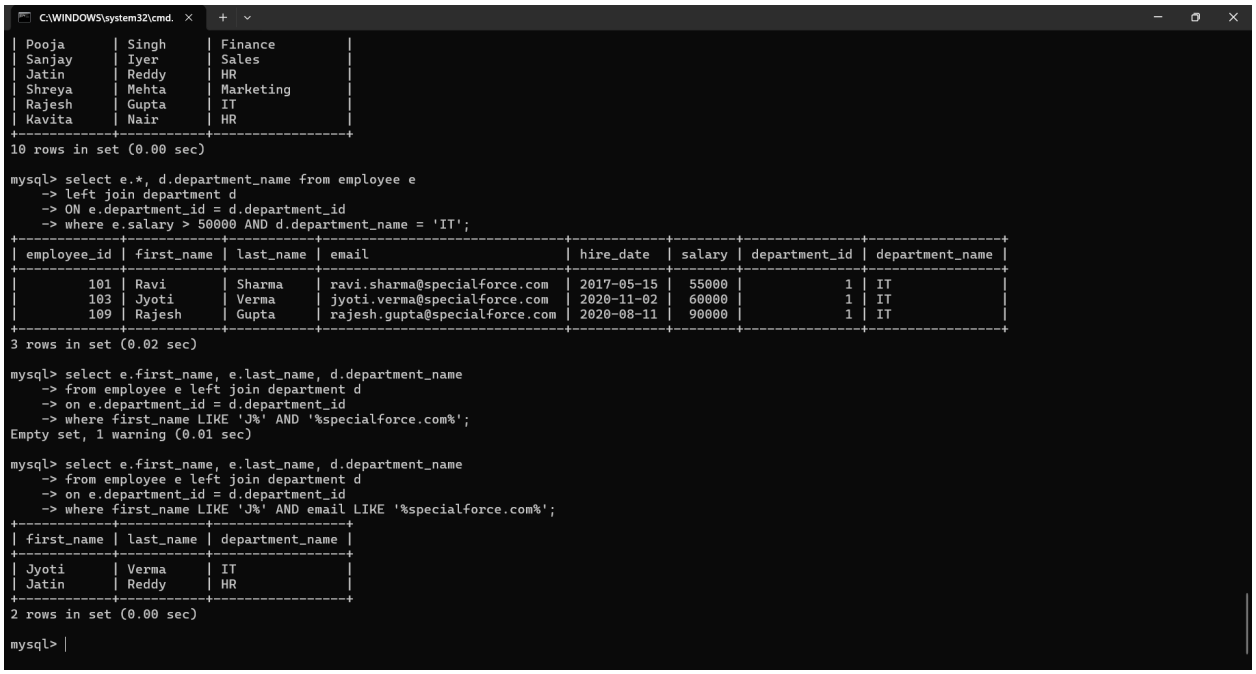
```
C:\WINDOWS\system32\cmd.

mysql> select e.first_name, e.last_name, d.department_name
    -> from employee e left join department d
    -> ON e.department_id = d.department_id
    -> where e.department_id is not null;
+------------+-----------+-----------------+
| first_name | last_name | department_name |
+------------+-----------+-----------------+
| Ravi       | Sharma    | IT              |
| Neha       | Kapoor    | HR              |
| Jyoti      | Verma     | IT              |
| Anil       | Patil     | Sales           |
| Pooja      | Singh     | Finance         |
| Sanjay     | Iyer      | Sales           |
| Jatin      | Reddy     | HR              |
| Shreya     | Mehta     | Marketing       |
| Rajesh     | Gupta     | IT              |
| Kavita     | Nair      | HR              |
+------------+-----------+-----------------+
10 rows in set (0.00 sec)

mysql> select e.*, d.department_name from employee e
    -> left join department d
    -> ON e.department_id = d.department_id
    -> where e.salary > 50000 AND d.department_name = 'IT';
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+-----------------+
| employee_id | first_name | last_name | email                         | hire_date  | salary | department_id | department_name |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+-----------------+
|         101 | Ravi       | Sharma    | ravi.sharma@specialforce.com  | 2017-05-15 |  55000 |             1 | IT              |
|         103 | Jyoti      | Verma     | jyoti.verma@specialforce.com  | 2020-11-02 |  60000 |             1 | IT              |
|         109 | Rajesh     | Gupta     | rajesh.gupta@specialforce.com | 2020-08-11 |  90000 |             1 | IT              |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+-----------------+
3 rows in set (0.02 sec)

mysql>
```

Query 3



```
C:\WINDOWS\system32\cmd.

| Pooja      | Singh     | Finance         |
| Sanjay     | Iyer      | Sales           |
| Jatin      | Reddy     | HR              |
| Shreya     | Mehta     | Marketing       |
| Rajesh     | Gupta     | IT              |
| Kavita     | Nair      | HR              |
+------------+-----------+-----------------+
10 rows in set (0.00 sec)

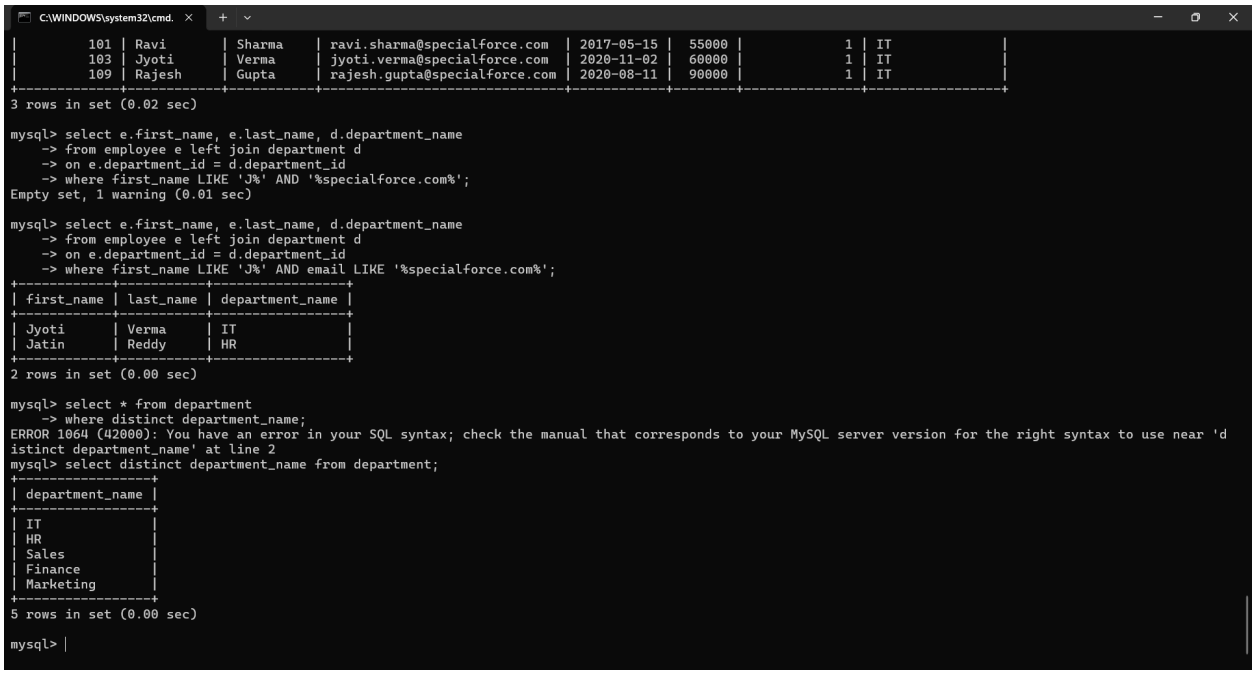
mysql> select e.*, d.department_name from employee e
    -> left join department d
    -> ON e.department_id = d.department_id
    -> where e.salary > 50000 AND d.department_name = 'IT';
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+-----------------+
| employee_id | first_name | last_name | email                         | hire_date  | salary | department_id | department_name |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+-----------------+
|         101 | Ravi       | Sharma    | ravi.sharma@specialforce.com  | 2017-05-15 |  55000 |             1 | IT              |
|         103 | Jyoti      | Verma     | jyoti.verma@specialforce.com  | 2020-11-02 |  60000 |             1 | IT              |
|         109 | Rajesh     | Gupta     | rajesh.gupta@specialforce.com | 2020-08-11 |  90000 |             1 | IT              |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+-----------------+
3 rows in set (0.02 sec)

mysql> select e.first_name, e.last_name, d.department_name
    -> from employee e left join department d
    -> on e.department_id = d.department_id
    -> where first_name LIKE 'J%' AND '%specialforce.com%';
Empty set, 1 warning (0.01 sec)

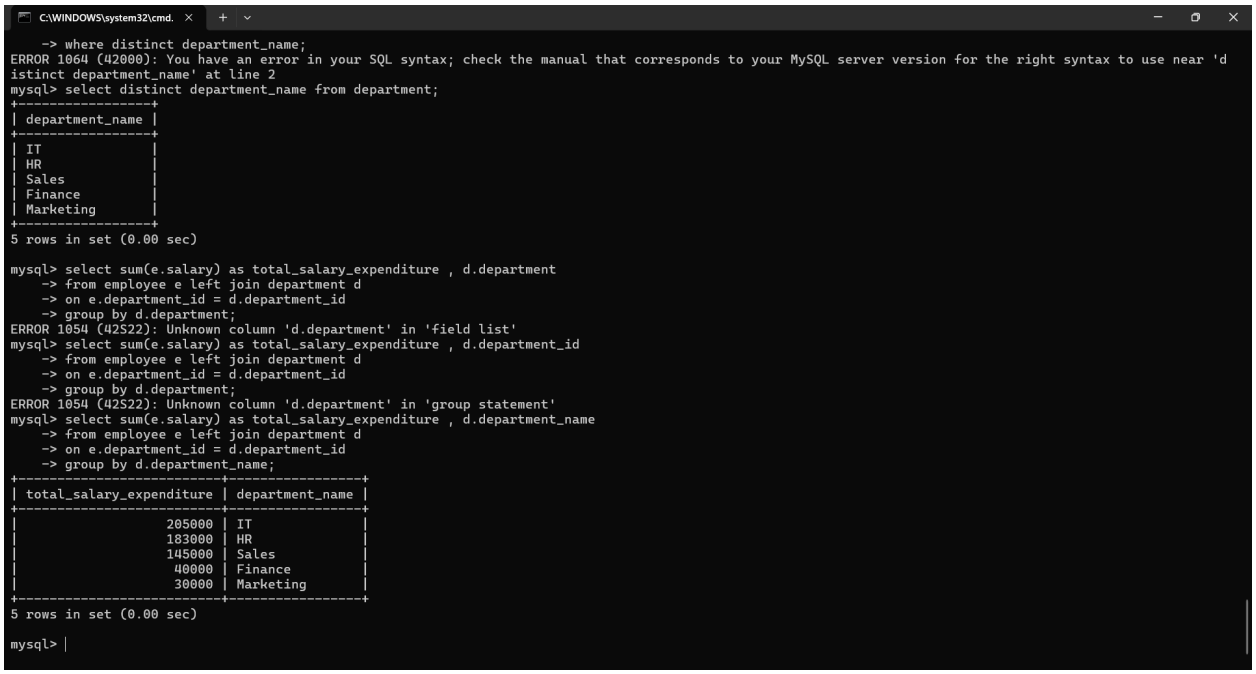
mysql> select e.first_name, e.last_name, d.department_name
    -> from employee e left join department d
    -> on e.department_id = d.department_id
    -> where first_name LIKE 'J%' AND email LIKE '%specialforce.com%';
+------------+-----------+-----------------+
| first_name | last_name | department_name |
+------------+-----------+-----------------+
| Jyoti      | Verma     | IT              |
| Jatin      | Reddy     | HR              |
+------------+-----------+-----------------+
2 rows in set (0.00 sec)

mysql>
```

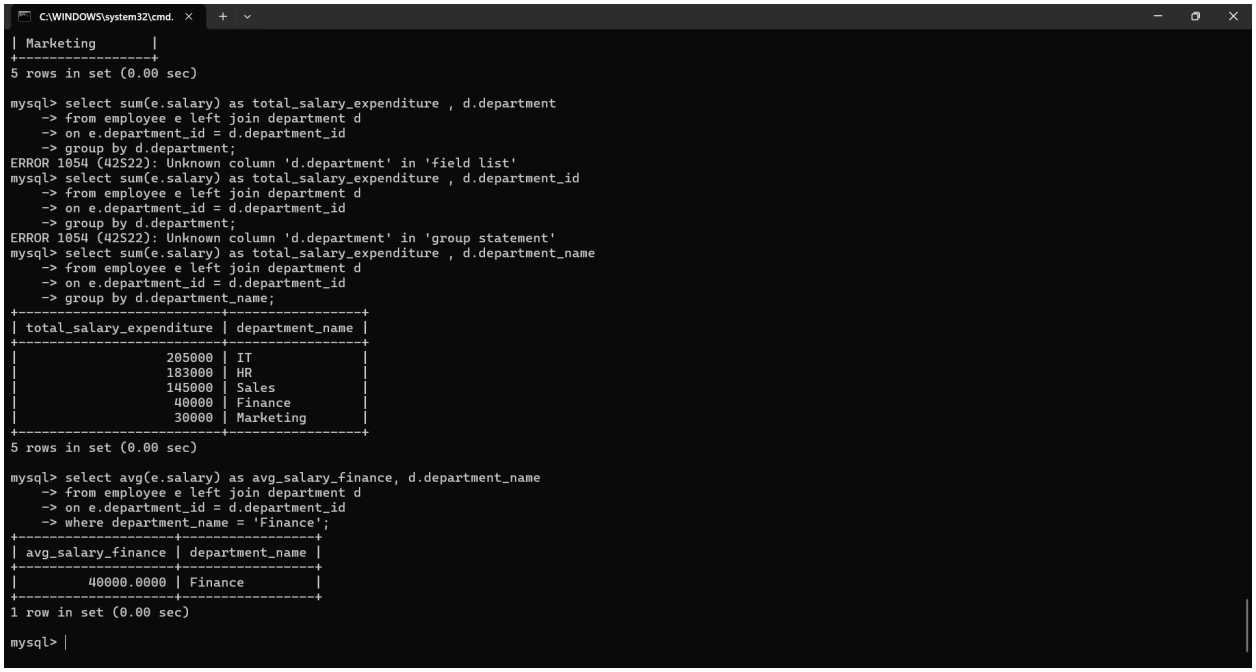
Query 4



Query 5



Query 6



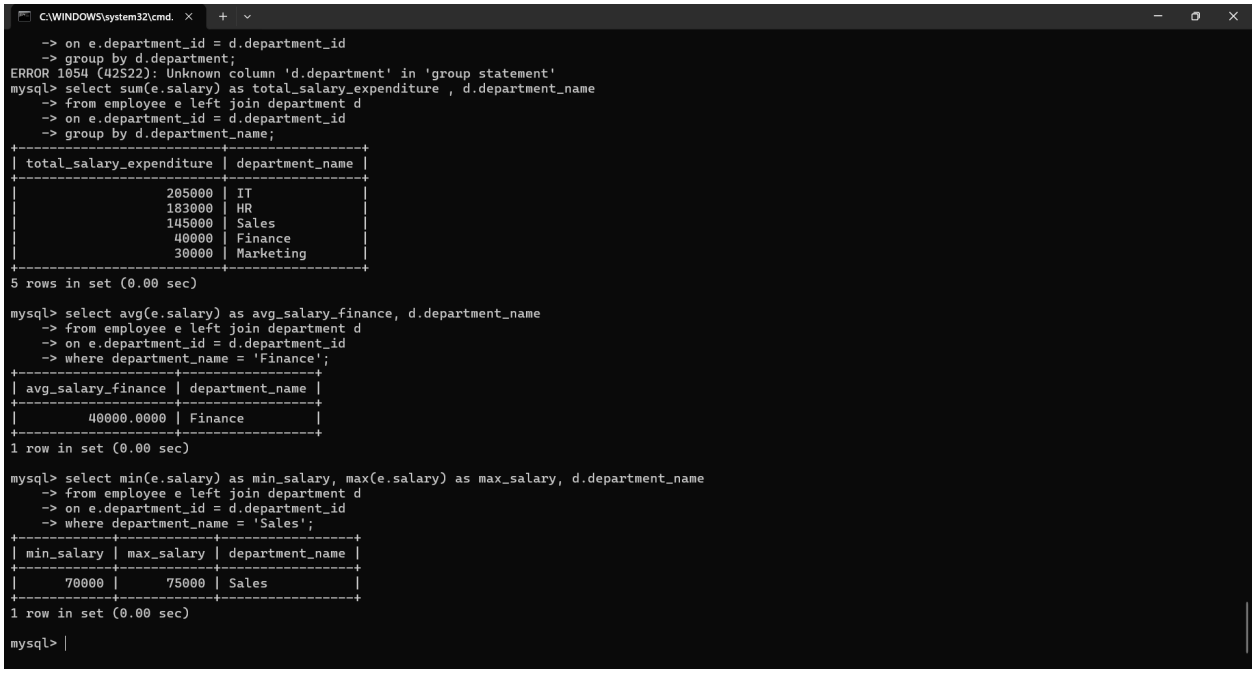
```
C:\WINDOWS\system32\cmd.
| Marketing       |
+-----------------+
5 rows in set (0.00 sec)

mysql> select sum(e.salary) as total_salary_expenditure , d.department
    -> from employee e left join department d
    -> on e.department_id = d.department_id
    -> group by d.department;
ERROR 1054 (42S22): Unknown column 'd.department' in 'field list'
mysql> select sum(e.salary) as total_salary_expenditure , d.department_id
    -> from employee e left join department d
    -> on e.department_id = d.department_id
    -> group by d.department;
ERROR 1054 (42S22): Unknown column 'd.department' in 'group statement'
mysql> select sum(e.salary) as total_salary_expenditure , d.department_name
    -> from employee e left join department d
    -> on e.department_id = d.department_id
    -> group by d.department_name;
+--------------------------+-----------------+
| total_salary_expenditure | department_name |
+--------------------------+-----------------+
|                   205000 | IT              |
|                   183000 | HR              |
|                   145000 | Sales           |
|                    40000 | Finance         |
|                    30000 | Marketing       |
+--------------------------+-----------------+
5 rows in set (0.00 sec)

mysql> select avg(e.salary) as avg_salary_finance, d.department_name
    -> from employee e left join department d
    -> on e.department_id = d.department_id
    -> where department_name = 'Finance';
+--------------------+-----------------+
| avg_salary_finance | department_name |
+--------------------+-----------------+
|         40000.0000 | Finance         |
+--------------------+-----------------+
1 row in set (0.00 sec)

mysql>
```

Query 7



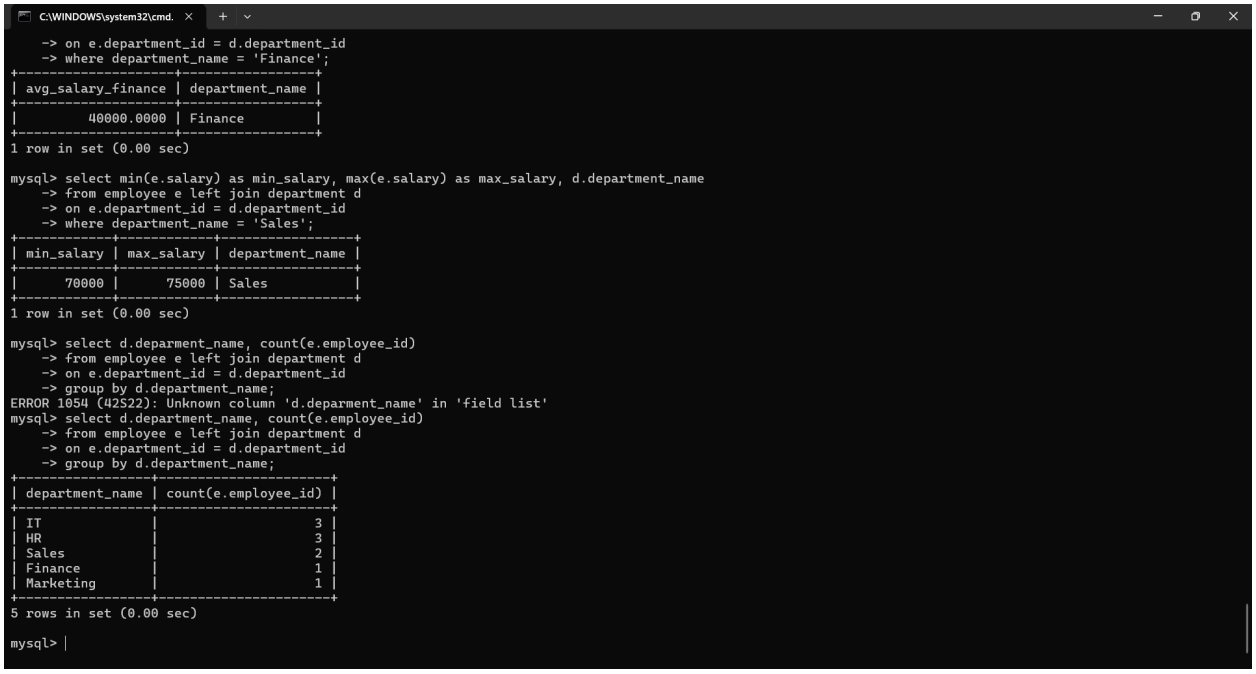
```
C:\WINDOWS\system32\cmd.
    -> on e.department_id = d.department_id
    -> group by d.department;
ERROR 1054 (42S22): Unknown column 'd.department' in 'group statement'
mysql> select sum(e.salary) as total_salary_expenditure , d.department_name
    -> from employee e left join department d
    -> on e.department_id = d.department_id
    -> group by d.department_name;
+--------------------------+-----------------+
| total_salary_expenditure | department_name |
+--------------------------+-----------------+
|                   205000 | IT              |
|                   183000 | HR              |
|                   145000 | Sales           |
|                    40000 | Finance         |
|                    30000 | Marketing       |
+--------------------------+-----------------+
5 rows in set (0.00 sec)

mysql> select avg(e.salary) as avg_salary_finance, d.department_name
    -> from employee e left join department d
    -> on e.department_id = d.department_id
    -> where department_name = 'Finance';
+--------------------+-----------------+
| avg_salary_finance | department_name |
+--------------------+-----------------+
|         40000.0000 | Finance         |
+--------------------+-----------------+
1 row in set (0.00 sec)

mysql> select min(e.salary) as min_salary, max(e.salary) as max_salary, d.department_name
    -> from employee e left join department d
    -> on e.department_id = d.department_id
    -> where department_name = 'Sales';
+------------+------------+-----------------+
| min_salary | max_salary | department_name |
+------------+------------+-----------------+
|      70000 |      75000 | Sales           |
+------------+------------+-----------------+
1 row in set (0.00 sec)

mysql>
```

Query 8



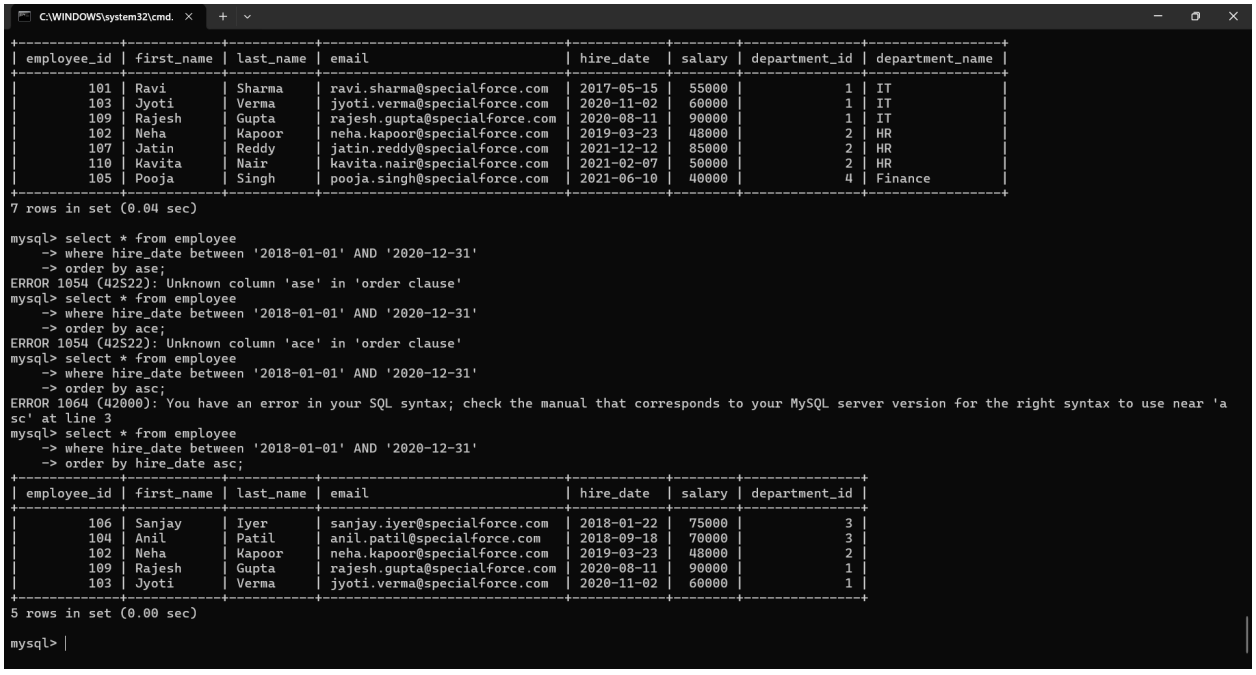
```
C:\WINDOWS\system32\cmd.
    -> on e.department_id = d.department_id
    -> where department_name = 'Finance';
+--------------------+-----------------+
| avg_salary_finance | department_name |
+--------------------+-----------------+
|         40000.0000 | Finance         |
+--------------------+-----------------+
1 row in set (0.00 sec)

mysql> select min(e.salary) as min_salary, max(e.salary) as max_salary, d.department_name
    -> from employee e left join department d
    -> on e.department_id = d.department_id
    -> where department_name = 'Sales';
+------------+------------+-----------------+
| min_salary | max_salary | department_name |
+------------+------------+-----------------+
|      70000 |      75000 | Sales           |
+------------+------------+-----------------+
1 row in set (0.00 sec)

mysql> select d.deparment_name, count(e.employee_id)
    -> from employee e left join department d
    -> on e.department_id = d.department_id
    -> group by d.department_name;
ERROR 1054 (42S22): Unknown column 'd.deparment_name' in 'field list'
mysql> select d.department_name, count(e.employee_id)
    -> from employee e left join department d
    -> on e.department_id = d.department_id
    -> group by d.department_name;
+-----------------+----------------------+
| department_name | count(e.employee_id) |
+-----------------+----------------------+
| IT              |                    3 |
| HR              |                    3 |
| Sales           |                    2 |
| Finance         |                    1 |
| Marketing       |                    1 |
+-----------------+----------------------+
5 rows in set (0.00 sec)

mysql>
```

Query 9

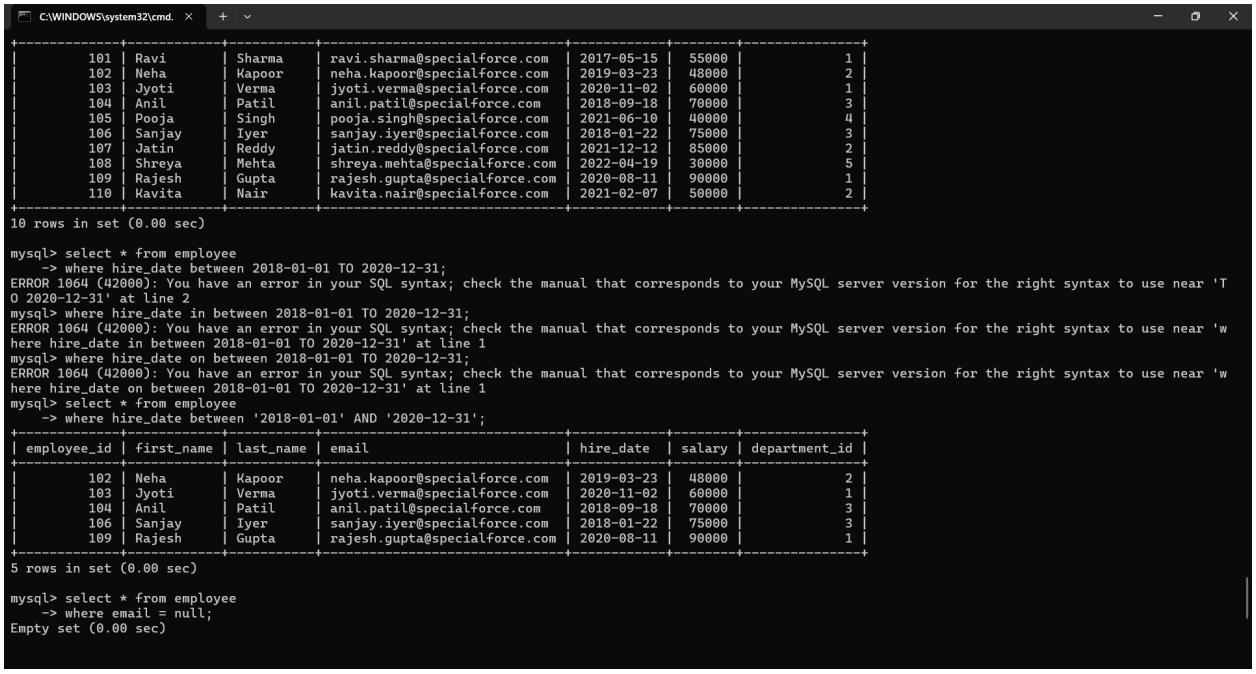


```
C:\WINDOWS\system32\cmd.
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+-----------------+
| employee_id | first_name | last_name | email                         | hire_date  | salary | department_id | department_name |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+-----------------+
|         101 | Ravi       | Sharma    | ravi.sharma@specialforce.com  | 2017-05-15 |  55000 |             1 | IT              |
|         103 | Jyoti      | Verma     | jyoti.verma@specialforce.com  | 2020-11-02 |  60000 |             1 | IT              |
|         109 | Rajesh     | Gupta     | rajesh.gupta@specialforce.com | 2020-08-11 |  90000 |             1 | IT              |
|         102 | Neha       | Kapoor    | neha.kapoor@specialforce.com  | 2019-03-23 |  48000 |             2 | HR              |
|         107 | Jatin      | Reddy     | jatin.reddy@specialforce.com  | 2021-12-12 |  85000 |             2 | HR              |
|         110 | Kavita     | Nair      | kavita.nair@specialforce.com  | 2021-02-07 |  50000 |             2 | HR              |
|         105 | Pooja      | Singh     | pooja.singh@specialforce.com  | 2021-06-10 |  40000 |             4 | Finance         |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+-----------------+
7 rows in set (0.04 sec)

mysql> select * from employee
    -> where hire_date between '2018-01-01' AND '2020-12-31'
    -> order by ase;
ERROR 1054 (42S22): Unknown column 'ase' in 'order clause'
mysql> select * from employee
    -> where hire_date between '2018-01-01' AND '2020-12-31'
    -> order by ace;
ERROR 1054 (42S22): Unknown column 'ace' in 'order clause'
mysql> select * from employee
    -> where hire_date between '2018-01-01' AND '2020-12-31'
    -> order by asc;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'a
sc' at line 3
mysql> select * from employee
    -> where hire_date between '2018-01-01' AND '2020-12-31'
    -> order by hire_date asc;
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+
| employee_id | first_name | last_name | email                         | hire_date  | salary | department_id |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+
|         106 | Sanjay     | Iyer      | sanjay.iyer@specialforce.com  | 2018-01-22 |  75000 |             3 |
|         104 | Anil       | Patil     | anil.patil@specialforce.com   | 2018-09-18 |  70000 |             3 |
|         102 | Neha       | Kapoor    | neha.kapoor@specialforce.com  | 2019-03-23 |  48000 |             2 |
|         109 | Rajesh     | Gupta     | rajesh.gupta@specialforce.com | 2020-08-11 |  90000 |             1 |
|         103 | Jyoti      | Verma     | jyoti.verma@specialforce.com  | 2020-11-02 |  60000 |             1 |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+
5 rows in set (0.00 sec)

mysql>
```

Query 10

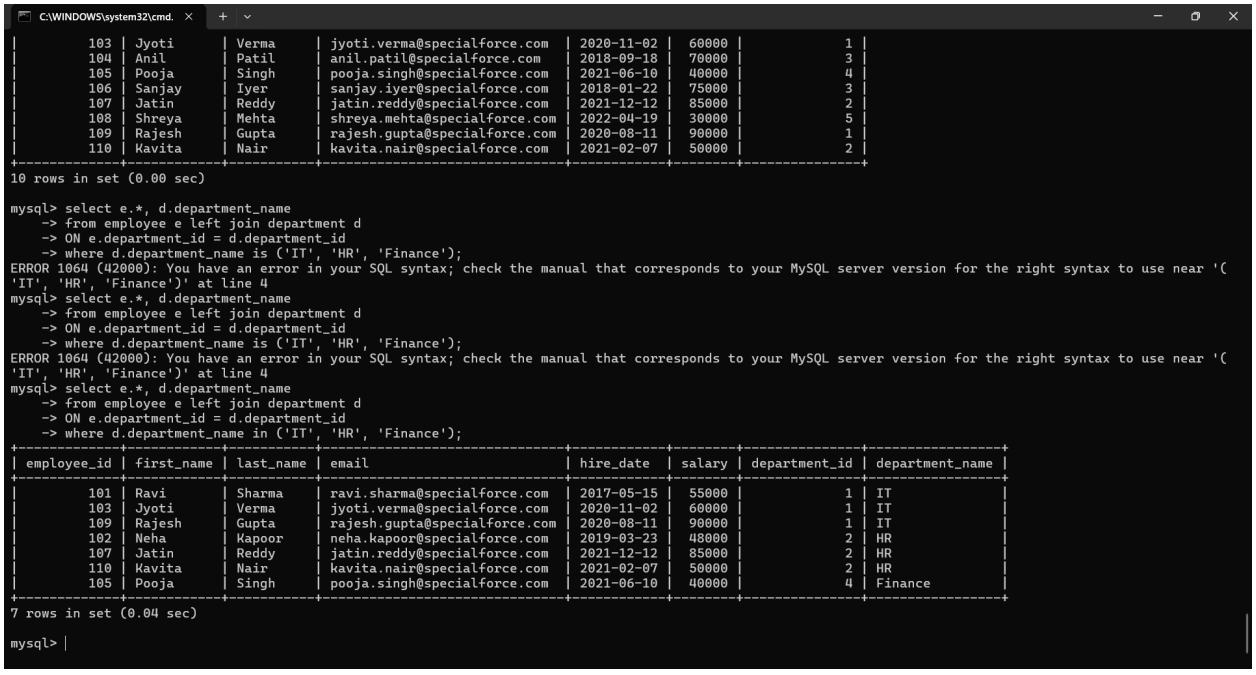


```
C:\WINDOWS\system32\cmd.
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+
|         101 | Ravi       | Sharma    | ravi.sharma@specialforce.com  | 2017-05-15 |  55000 |             1 |
|         102 | Neha       | Kapoor    | neha.kapoor@specialforce.com  | 2019-03-23 |  48000 |             2 |
|         103 | Jyoti      | Verma     | jyoti.verma@specialforce.com  | 2020-11-02 |  60000 |             1 |
|         104 | Anil       | Patil     | anil.patil@specialforce.com   | 2018-09-18 |  70000 |             3 |
|         105 | Pooja      | Singh     | pooja.singh@specialforce.com  | 2021-06-10 |  40000 |             4 |
|         106 | Sanjay     | Iyer      | sanjay.iyer@specialforce.com  | 2018-01-22 |  75000 |             3 |
|         107 | Jatin      | Reddy     | jatin.reddy@specialforce.com  | 2021-12-12 |  85000 |             2 |
|         108 | Shreya     | Mehta     | shreya.mehta@specialforce.com | 2022-04-19 |  30000 |             5 |
|         109 | Rajesh     | Gupta     | rajesh.gupta@specialforce.com | 2020-08-11 |  90000 |             1 |
|         110 | Kavita     | Nair      | kavita.nair@specialforce.com  | 2021-02-07 |  50000 |             2 |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+
10 rows in set (0.00 sec)

mysql> select * from employee
    -> where hire_date between 2018-01-01 TO 2020-12-31;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'T
O 2020-12-31' at line 2
mysql> where hire_date in between 2018-01-01 TO 2020-12-31;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'w
here hire_date in between 2018-01-01 TO 2020-12-31' at line 1
mysql> where hire_date on between 2018-01-01 TO 2020-12-31;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'w
here hire_date on between 2018-01-01 TO 2020-12-31' at line 1
mysql> select * from employee
    -> where hire_date between '2018-01-01' AND '2020-12-31';
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+
| employee_id | first_name | last_name | email                         | hire_date  | salary | department_id |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+
|         102 | Neha       | Kapoor    | neha.kapoor@specialforce.com  | 2019-03-23 |  48000 |             2 |
|         103 | Jyoti      | Verma     | jyoti.verma@specialforce.com  | 2020-11-02 |  60000 |             1 |
|         104 | Anil       | Patil     | anil.patil@specialforce.com   | 2018-09-18 |  70000 |             3 |
|         106 | Sanjay     | Iyer      | sanjay.iyer@specialforce.com  | 2018-01-22 |  75000 |             3 |
|         109 | Rajesh     | Gupta     | rajesh.gupta@specialforce.com | 2020-08-11 |  90000 |             1 |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+
5 rows in set (0.00 sec)

mysql> select * from employee
    -> where email = null;
Empty set (0.00 sec)
```

Query 11



```
C:\WINDOWS\system32\cmd.
|         103 | Jyoti      | Verma     | jyoti.verma@specialforce.com  | 2020-11-02 |  60000 |             1 |
|         104 | Anil       | Patil     | anil.patil@specialforce.com   | 2018-09-18 |  70000 |             3 |
|         105 | Pooja      | Singh     | pooja.singh@specialforce.com  | 2021-06-10 |  40000 |             4 |
|         106 | Sanjay     | Iyer      | sanjay.iyer@specialforce.com  | 2018-01-22 |  75000 |             3 |
|         107 | Jatin      | Reddy     | jatin.reddy@specialforce.com  | 2021-12-12 |  85000 |             2 |
|         108 | Shreya     | Mehta     | shreya.mehta@specialforce.com | 2022-04-19 |  30000 |             5 |
|         109 | Rajesh     | Gupta     | rajesh.gupta@specialforce.com | 2020-08-11 |  90000 |             1 |
|         110 | Kavita     | Nair      | kavita.nair@specialforce.com  | 2021-02-07 |  50000 |             2 |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+
10 rows in set (0.00 sec)

mysql> select e.*, d.department_name
    -> from employee e left join department d
    -> ON e.department_id = d.department_id
    -> where d.department_name is ('IT', 'HR', 'Finance');
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '(
'IT', 'HR', 'Finance')' at line 4
mysql> select e.*, d.department_name
    -> from employee e left join department d
    -> ON e.department_id = d.department_id
    -> where d.department_name is ('IT', 'HR', 'Finance');
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '(
'IT', 'HR', 'Finance')' at line 4
mysql> select e.*, d.department_name
    -> from employee e left join department d
    -> ON e.department_id = d.department_id
    -> where d.department_name in ('IT', 'HR', 'Finance');
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+-----------------+
| employee_id | first_name | last_name | email                         | hire_date  | salary | department_id | department_name |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+-----------------+
|         101 | Ravi       | Sharma    | ravi.sharma@specialforce.com  | 2017-05-15 |  55000 |             1 | IT              |
|         103 | Jyoti      | Verma     | jyoti.verma@specialforce.com  | 2020-11-02 |  60000 |             1 | IT              |
|         109 | Rajesh     | Gupta     | rajesh.gupta@specialforce.com | 2020-08-11 |  90000 |             1 | IT              |
|         102 | Neha       | Kapoor    | neha.kapoor@specialforce.com  | 2019-03-23 |  48000 |             2 | HR              |
|         107 | Jatin      | Reddy     | jatin.reddy@specialforce.com  | 2021-12-12 |  85000 |             2 | HR              |
|         110 | Kavita     | Nair      | kavita.nair@specialforce.com  | 2021-02-07 |  50000 |             2 | HR              |
|         105 | Pooja      | Singh     | pooja.singh@specialforce.com  | 2021-06-10 |  40000 |             4 | Finance         |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+-----------------+
7 rows in set (0.04 sec)

mysql>
```

Query 12

```
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+
|         106 | Sanjay     | Iyer      | sanjay.iyer@specialforce.com  | 2018-01-22 |  75000 |             3 |
|         104 | Anil       | Patil     | anil.patil@specialforce.com   | 2018-09-18 |  70000 |             3 |
|         102 | Neha       | Kapoor    | neha.kapoor@specialforce.com  | 2019-03-23 |  48000 |             2 |
|         109 | Rajesh     | Gupta     | rajesh.gupta@specialforce.com | 2020-08-11 |  90000 |             1 |
|         103 | Jyoti      | Verma     | jyoti.verma@specialforce.com  | 2020-11-02 |  60000 |             1 |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+
5 rows in set (0.00 sec)

mysql> select first_name, last_name, salary from employee
    -> where salary between 30000 AND 70000;
+------------+-----------+--------+
| first_name | last_name | salary |
+------------+-----------+--------+
| Ravi       | Sharma    |  55000 |
| Neha       | Kapoor    |  48000 |
| Jyoti      | Verma     |  60000 |
| Anil       | Patil     |  70000 |
| Pooja      | Singh     |  40000 |
| Shreya     | Mehta     |  30000 |
| Kavita     | Nair      |  50000 |
+------------+-----------+--------+
7 rows in set (0.00 sec)

mysql> select first_name, last_name, salary from employee
    -> where salary between 30000 AND 70000
    -> order by salary desc;
+------------+-----------+--------+
| first_name | last_name | salary |
+------------+-----------+--------+
| Anil       | Patil     |  70000 |
| Jyoti      | Verma     |  60000 |
| Ravi       | Sharma    |  55000 |
| Kavita     | Nair      |  50000 |
| Neha       | Kapoor    |  48000 |
| Pooja      | Singh     |  40000 |
| Shreya     | Mehta     |  30000 |
+------------+-----------+--------+
7 rows in set (0.00 sec)

mysql>
```

TMT

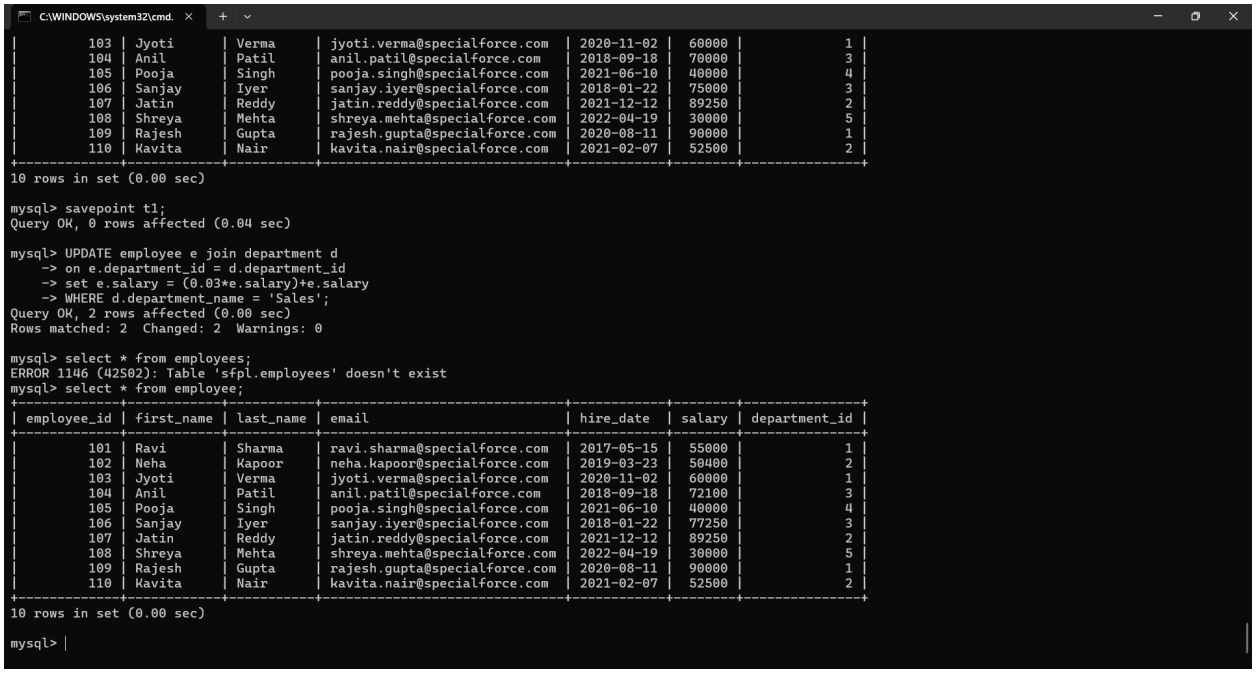
Task1

```
on e.department_id = d.department_id
set e.salary = (0.05*e.sal' at line 2
mysql> select e.*, d.department_name
    -> from employee e left join department d
    -> on e.department_id = d.department_id
    -> set e.salary = (0.05*e.salary)+e.salary
    -> in d.department_name = 'HR';
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 's
et e.salary = (0.05*e.salary)+e.salary
in d.department_name = 'HR'' at line 4
mysql> UPDATE employee e join department d
    -> on e.department_id = d.department_id
    -> SET e.salary = (0.05*e.salary)+e.salary
    -> in d.department_name = 'HR';
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'd
.department_name = 'HR'' at line 4
mysql>  UPDATE employee e join department d
    -> on e.department_id = d.department_id
    -> SET e.salary = (0.05*e.salary)+e.salary
    -> WHERE d.department_name = 'HR';
Query OK, 3 rows affected (0.04 sec)
Rows matched: 3  Changed: 3  Warnings: 0

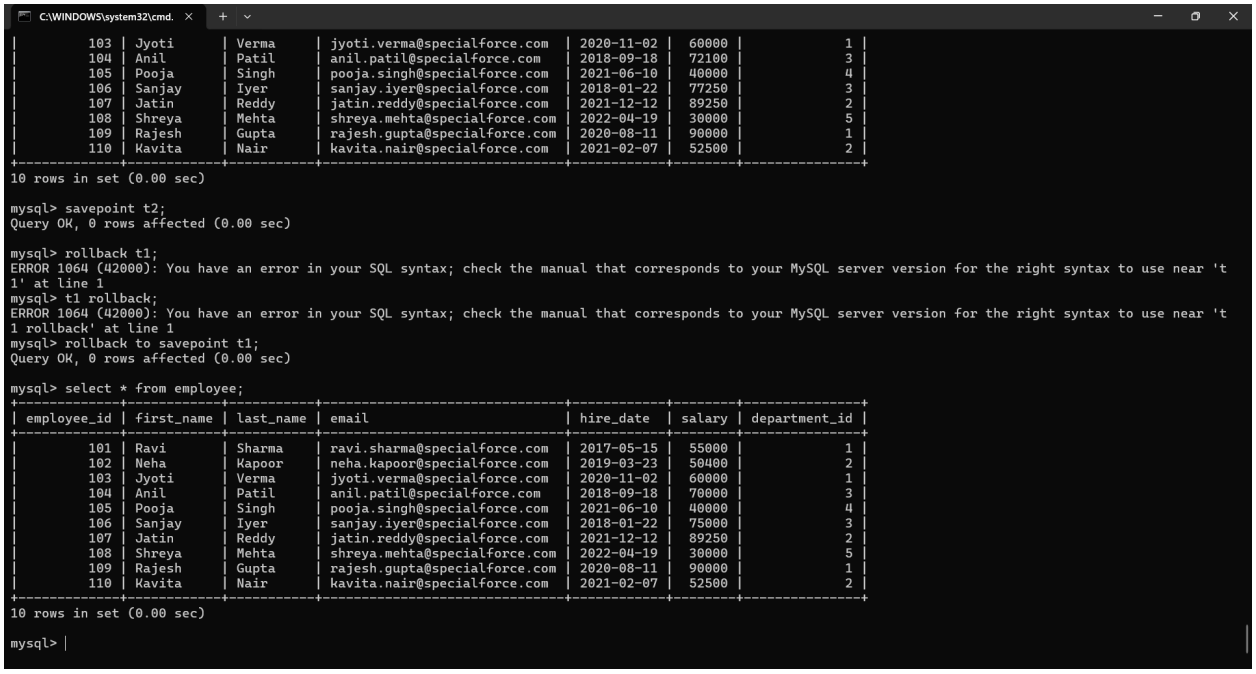
mysql>  select * from employee;
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+
| employee_id | first_name | last_name | email                         | hire_date  | salary | department_id |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+
|         101 | Ravi       | Sharma    | ravi.sharma@specialforce.com  | 2017-05-15 |  55000 |             1 |
|         102 | Neha       | Kapoor    | neha.kapoor@specialforce.com  | 2019-03-23 |  50400 |             2 |
|         103 | Jyoti      | Verma     | jyoti.verma@specialforce.com  | 2020-11-02 |  60000 |             1 |
|         104 | Anil       | Patil     | anil.patil@specialforce.com   | 2018-09-18 |  70000 |             3 |
|         105 | Pooja      | Singh     | pooja.singh@specialforce.com  | 2021-06-10 |  40000 |             4 |
|         106 | Sanjay     | Iyer      | sanjay.iyer@specialforce.com  | 2018-01-22 |  75000 |             3 |
|         107 | Jatin      | Reddy     | jatin.reddy@specialforce.com  | 2021-12-12 |  89250 |             2 |
|         108 | Shreya     | Mehta     | shreya.mehta@specialforce.com | 2022-04-19 |  30000 |             5 |
|         109 | Rajesh     | Gupta     | rajesh.gupta@specialforce.com | 2020-08-11 |  90000 |             1 |
|         110 | Kavita     | Nair      | kavita.nair@specialforce.com  | 2021-02-07 |  52500 |             2 |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+
10 rows in set (0.00 sec)

mysql>
```

Task 2


C:\WINDOWS\system32\cmd.
103	Jyoti	Verma	jyoti.verma@specialforce.com	2020-11-02	60000	1
104	Anil	Patil	anil.patil@specialforce.com	2018-09-18	70000	3
105	Pooja	Singh	pooja.singh@specialforce.com	2021-06-10	40000	4
106	Sanjay	Iyer	sanjay.iyer@specialforce.com	2018-01-22	75000	3
107	Jatin	Reddy	jatin.reddy@specialforce.com	2021-12-12	89250	2
108	Shreya	Mehta	shreya.mehta@specialforce.com	2022-04-19	30000	5
109	Rajesh	Gupta	rajesh.gupta@specialforce.com	2020-08-11	90000	1
110	Kavita	Nair	kavita.nair@specialforce.com	2021-02-07	52500	2
10 rows in set (0.00 sec)
mysql> savepoint t1;
Query OK, 0 rows affected (0.04 sec)
mysql> UPDATE employee e join department d
-> on e.department_id = d.department_id
-> set e.salary = (0.03*e.salary)+e.salary
-> WHERE d.department_name = 'Sales';
Query OK, 2 rows affected (0.00 sec)
Rows matched: 2 Changed: 2 Warnings: 0
mysql> select * from employees;
ERROR 1146 (42S02): Table 'sfpl.employees' doesn't exist
mysql> select * from employee;
employee_id	first_name	last_name	email	hire_date	salary	department_id
101	Ravi	Sharma	ravi.sharma@specialforce.com	2017-05-15	55000	1
102	Neha	Kapoor	neha.kapoor@specialforce.com	2019-03-23	50400	2
103	Jyoti	Verma	jyoti.verma@specialforce.com	2020-11-02	60000	1
104	Anil	Patil	anil.patil@specialforce.com	2018-09-18	72100	3
105	Pooja	Singh	pooja.singh@specialforce.com	2021-06-10	40000	4
106	Sanjay	Iyer	sanjay.iyer@specialforce.com	2018-01-22	77250	3
107	Jatin	Reddy	jatin.reddy@specialforce.com	2021-12-12	89250	2
108	Shreya	Mehta	shreya.mehta@specialforce.com	2022-04-19	30000	5
109	Rajesh	Gupta	rajesh.gupta@specialforce.com	2020-08-11	90000	1
110	Kavita	Nair	kavita.nair@specialforce.com	2021-02-07	52500	2
10 rows in set (0.00 sec)
mysql>

Task 3


C:\WINDOWS\system32\cmd.
103	Jyoti	Verma	jyoti.verma@specialforce.com	2020-11-02	60000	1
104	Anil	Patil	anil.patil@specialforce.com	2018-09-18	72100	3
105	Pooja	Singh	pooja.singh@specialforce.com	2021-06-10	40000	4
106	Sanjay	Iyer	sanjay.iyer@specialforce.com	2018-01-22	77250	3
107	Jatin	Reddy	jatin.reddy@specialforce.com	2021-12-12	89250	2
108	Shreya	Mehta	shreya.mehta@specialforce.com	2022-04-19	30000	5
109	Rajesh	Gupta	rajesh.gupta@specialforce.com	2020-08-11	90000	1
110	Kavita	Nair	kavita.nair@specialforce.com	2021-02-07	52500	2
10 rows in set (0.00 sec)
mysql> savepoint t2;
Query OK, 0 rows affected (0.00 sec)
mysql> rollback t1;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 't
1' at line 1
mysql> t1 rollback;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 't
1 rollback' at line 1
mysql> rollback to savepoint t1;
Query OK, 0 rows affected (0.00 sec)
mysql> select * from employee;
employee_id	first_name	last_name	email	hire_date	salary	department_id
101	Ravi	Sharma	ravi.sharma@specialforce.com	2017-05-15	55000	1
102	Neha	Kapoor	neha.kapoor@specialforce.com	2019-03-23	50400	2
103	Jyoti	Verma	jyoti.verma@specialforce.com	2020-11-02	60000	1
104	Anil	Patil	anil.patil@specialforce.com	2018-09-18	70000	3
105	Pooja	Singh	pooja.singh@specialforce.com	2021-06-10	40000	4
106	Sanjay	Iyer	sanjay.iyer@specialforce.com	2018-01-22	75000	3
107	Jatin	Reddy	jatin.reddy@specialforce.com	2021-12-12	89250	2
108	Shreya	Mehta	shreya.mehta@specialforce.com	2022-04-19	30000	5
109	Rajesh	Gupta	rajesh.gupta@specialforce.com	2020-08-11	90000	1
110	Kavita	Nair	kavita.nair@specialforce.com	2021-02-07	52500	2
10 rows in set (0.00 sec)
mysql>

Task 4

```
C:\WINDOWS\system32\cmd.
|         106 | Sanjay     | Iyer      | sanjay.iyer@specialforce.com  | 2018-01-22 |  77250 |             3 |
|         107 | Jatin      | Reddy     | jatin.reddy@specialforce.com  | 2021-12-12 |  89250 |             2 |
|         108 | Shreya     | Mehta     | shreya.mehta@specialforce.com | 2022-04-19 |  30000 |             5 |
|         109 | Rajesh     | Gupta     | rajesh.gupta@specialforce.com | 2020-08-11 |  90000 |             1 |
|         110 | Kavita     | Nair      | kavita.nair@specialforce.com  | 2021-02-07 |  52500 |             2 |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+
10 rows in set (0.00 sec)

mysql> savepoint t2;
Query OK, 0 rows affected (0.00 sec)

mysql> rollback t1;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 't
1' at line 1
mysql> t1 rollback;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 't
1 rollback' at line 1
mysql> rollback to savepoint t1;
Query OK, 0 rows affected (0.00 sec)

mysql> select * from employee;
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+
| employee_id | first_name | last_name | email                         | hire_date  | salary | department_id |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+
|         101 | Ravi       | Sharma    | ravi.sharma@specialforce.com  | 2017-05-15 |  55000 |             1 |
|         102 | Neha       | Kapoor    | neha.kapoor@specialforce.com  | 2019-03-23 |  50400 |             2 |
|         103 | Jyoti      | Verma     | jyoti.verma@specialforce.com  | 2020-11-02 |  60000 |             1 |
|         104 | Anil       | Patil     | anil.patil@specialforce.com   | 2018-09-18 |  70000 |             3 |
|         105 | Pooja      | Singh     | pooja.singh@specialforce.com  | 2021-06-10 |  40000 |             4 |
|         106 | Sanjay     | Iyer      | sanjay.iyer@specialforce.com  | 2018-01-22 |  75000 |             3 |
|         107 | Jatin      | Reddy     | jatin.reddy@specialforce.com  | 2021-12-12 |  89250 |             2 |
|         108 | Shreya     | Mehta     | shreya.mehta@specialforce.com | 2022-04-19 |  30000 |             5 |
|         109 | Rajesh     | Gupta     | rajesh.gupta@specialforce.com | 2020-08-11 |  90000 |             1 |
|         110 | Kavita     | Nair      | kavita.nair@specialforce.com  | 2021-02-07 |  52500 |             2 |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+
10 rows in set (0.00 sec)

mysql> commit;
Query OK, 0 rows affected (0.05 sec)

mysql>
```

Query 13

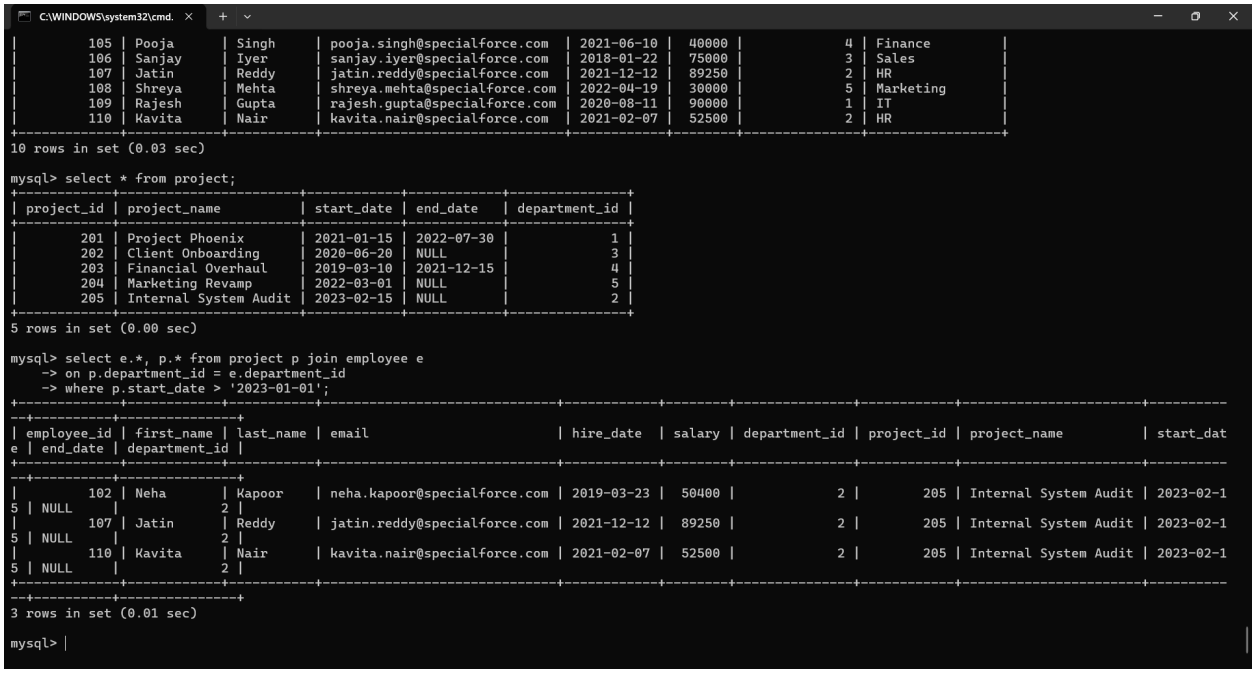
```
C:\WINDOWS\system32\cmd.
| employee_id | first_name | last_name | email                         | hire_date  | salary | department_id |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+
|         101 | Ravi       | Sharma    | ravi.sharma@specialforce.com  | 2017-05-15 |  55000 |             1 |
|         102 | Neha       | Kapoor    | neha.kapoor@specialforce.com  | 2019-03-23 |  50400 |             2 |
|         103 | Jyoti      | Verma     | jyoti.verma@specialforce.com  | 2020-11-02 |  60000 |             1 |
|         104 | Anil       | Patil     | anil.patil@specialforce.com   | 2018-09-18 |  70000 |             3 |
|         105 | Pooja      | Singh     | pooja.singh@specialforce.com  | 2021-06-10 |  40000 |             4 |
|         106 | Sanjay     | Iyer      | sanjay.iyer@specialforce.com  | 2018-01-22 |  75000 |             3 |
|         107 | Jatin      | Reddy     | jatin.reddy@specialforce.com  | 2021-12-12 |  89250 |             2 |
|         108 | Shreya     | Mehta     | shreya.mehta@specialforce.com | 2022-04-19 |  30000 |             5 |
|         109 | Rajesh     | Gupta     | rajesh.gupta@specialforce.com | 2020-08-11 |  90000 |             1 |
|         110 | Kavita     | Nair      | kavita.nair@specialforce.com  | 2021-02-07 |  52500 |             2 |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+
10 rows in set (0.00 sec)

mysql> commit;
Query OK, 0 rows affected (0.05 sec)

mysql> Write a query to join the Employees and Departments tables to list employees and their department names. Make sure all employees are included, even i
f they don’t belong to any department.^Z^C
mysql>
mysql> select e.*, d.department_name
    -> from employee e left join department d
    -> on e.department_id = d.department_id;
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+-----------------+
| employee_id | first_name | last_name | email                         | hire_date  | salary | department_id | department_name |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+-----------------+
|         101 | Ravi       | Sharma    | ravi.sharma@specialforce.com  | 2017-05-15 |  55000 |             1 | IT              |
|         102 | Neha       | Kapoor    | neha.kapoor@specialforce.com  | 2019-03-23 |  50400 |             2 | HR              |
|         103 | Jyoti      | Verma     | jyoti.verma@specialforce.com  | 2020-11-02 |  60000 |             1 | IT              |
|         104 | Anil       | Patil     | anil.patil@specialforce.com   | 2018-09-18 |  70000 |             3 | Sales           |
|         105 | Pooja      | Singh     | pooja.singh@specialforce.com  | 2021-06-10 |  40000 |             4 | Finance         |
|         106 | Sanjay     | Iyer      | sanjay.iyer@specialforce.com  | 2018-01-22 |  75000 |             3 | Sales           |
|         107 | Jatin      | Reddy     | jatin.reddy@specialforce.com  | 2021-12-12 |  89250 |             2 | HR              |
|         108 | Shreya     | Mehta     | shreya.mehta@specialforce.com | 2022-04-19 |  30000 |             5 | Marketing       |
|         109 | Rajesh     | Gupta     | rajesh.gupta@specialforce.com | 2020-08-11 |  90000 |             1 | IT              |
|         110 | Kavita     | Nair      | kavita.nair@specialforce.com  | 2021-02-07 |  52500 |             2 | HR              |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+-----------------+
10 rows in set (0.03 sec)

mysql>
```

Query 14



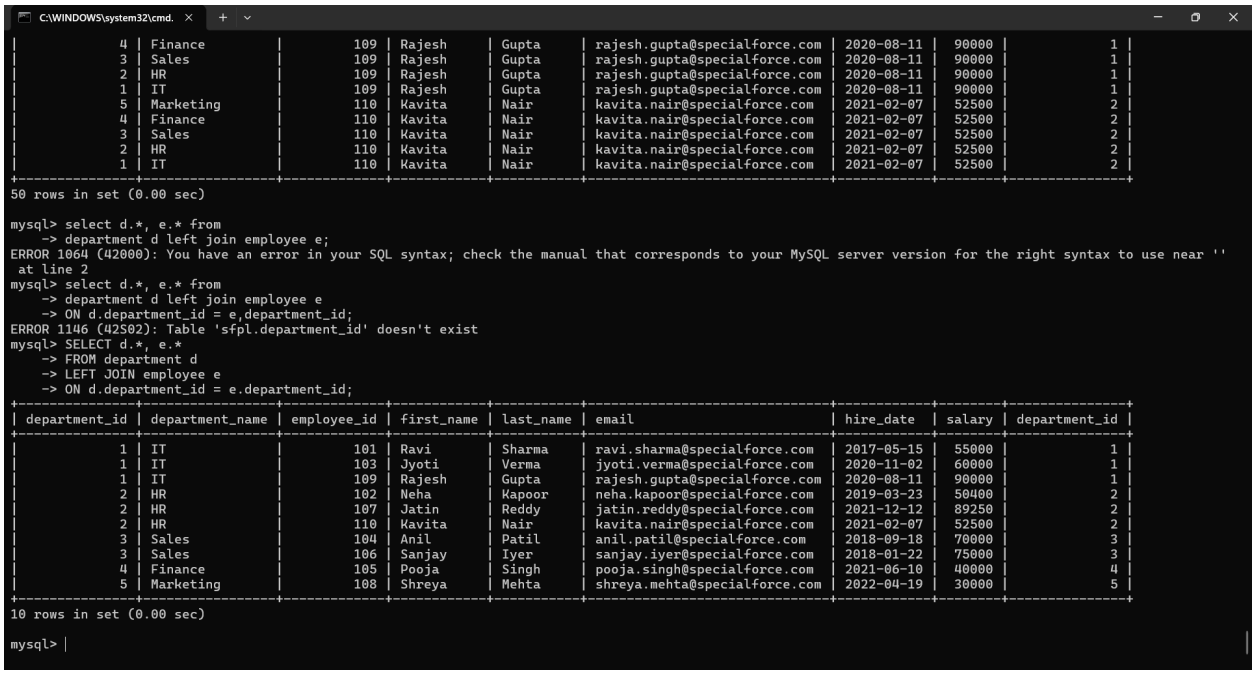
```
|         105 | Pooja      | Singh     | pooja.singh@specialforce.com   | 2021-06-10 |  40000 |             4 | Finance         |
|         106 | Sanjay     | Iyer      | sanjay.iyer@specialforce.com   | 2018-01-22 |  75000 |             3 | Sales           |
|         107 | Jatin      | Reddy     | jatin.reddy@specialforce.com   | 2021-12-12 |  89250 |             2 | HR              |
|         108 | Shreya     | Mehta     | shreya.mehta@specialforce.com  | 2022-04-19 |  30000 |             5 | Marketing       |
|         109 | Rajesh     | Gupta     | rajesh.gupta@specialforce.com  | 2020-08-11 |  90000 |             1 | IT              |
|         110 | Kavita     | Nair      | kavita.nair@specialforce.com   | 2021-02-07 |  52500 |             2 | HR              |
+-------------+------------+-----------+--------------------------------+------------+--------+---------------+-----------------+
10 rows in set (0.03 sec)

mysql> select * from project;
+------------+----------------------+------------+------------+---------------+
| project_id | project_name         | start_date | end_date   | department_id |
+------------+----------------------+------------+------------+---------------+
|        201 | Project Phoenix      | 2021-01-15 | 2022-07-30 |             1 |
|        202 | Client Onboarding    | 2020-06-20 | NULL       |             3 |
|        203 | Financial Overhaul   | 2019-03-10 | 2021-12-15 |             4 |
|        204 | Marketing Revamp     | 2022-03-01 | NULL       |             5 |
|        205 | Internal System Audit | 2023-02-15 | NULL       |             2 |
+------------+----------------------+------------+------------+---------------+
5 rows in set (0.00 sec)

mysql> select e.*, p.* from project p join employee e
    -> on p.department_id = e.department_id
    -> where p.start_date > '2023-01-01';
+-------------+------------+-----------+--------------------------------+------------+--------+---------------+------------+----------------------+-----------
--+----------+---------------+
| employee_id | first_name | last_name | email                          | hire_date  | salary | department_id | project_id | project_name         | start_dat
e | end_date | department_id |
+-------------+------------+-----------+--------------------------------+------------+--------+---------------+------------+----------------------+-----------
--+----------+---------------+
|         102 | Neha       | Kapoor    | neha.kapoor@specialforce.com | 2019-03-23 |  50400 |             2 |        205 | Internal System Audit | 2023-02-1
5 | NULL     |             2 |
|         107 | Jatin      | Reddy     | jatin.reddy@specialforce.com | 2021-12-12 |  89250 |             2 |        205 | Internal System Audit | 2023-02-1
5 | NULL     |             2 |
|         110 | Kavita     | Nair      | kavita.nair@specialforce.com | 2021-02-07 |  52500 |             2 |        205 | Internal System Audit | 2023-02-1
5 | NULL     |             2 |
+-------------+------------+-----------+--------------------------------+------------+--------+---------------+------------+----------------------+-----------
--+----------+---------------+
3 rows in set (0.01 sec)

mysql>
```

Query 15

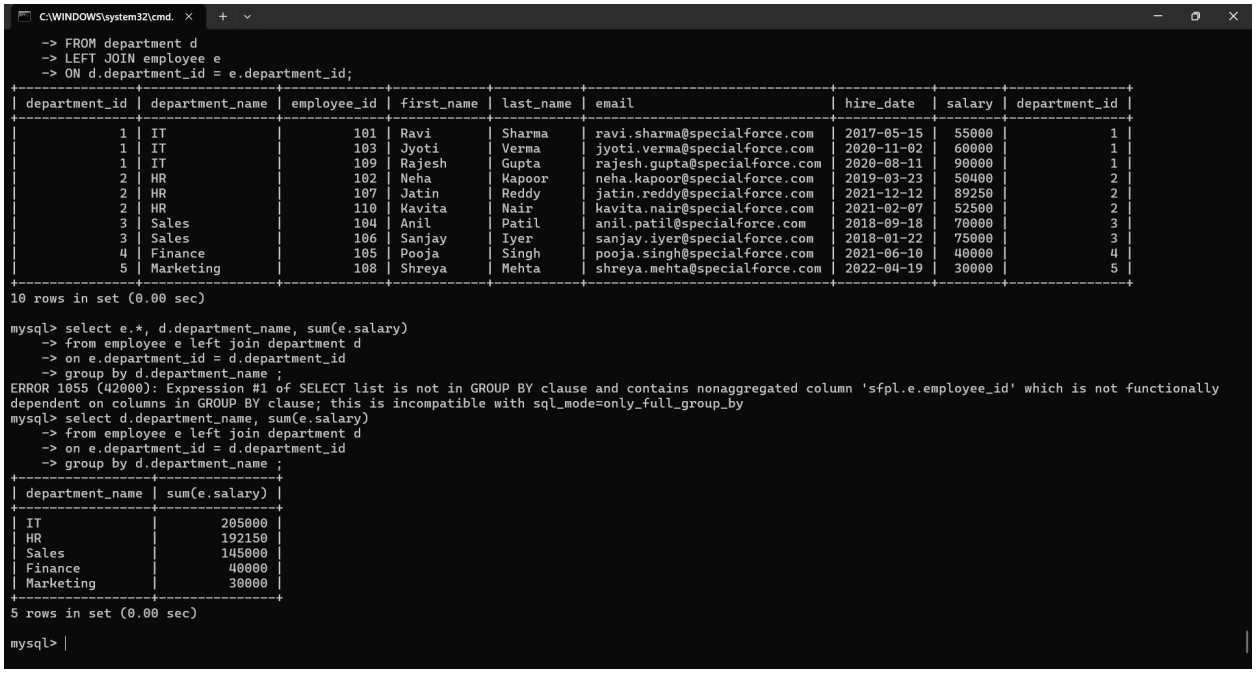


```
|             4 | Finance          |         109 | Rajesh     | Gupta     | rajesh.gupta@specialforce.com | 2020-08-11 |  90000 |             1 |
|             3 | Sales            |         109 | Rajesh     | Gupta     | rajesh.gupta@specialforce.com | 2020-08-11 |  90000 |             1 |
|             2 | HR               |         109 | Rajesh     | Gupta     | rajesh.gupta@specialforce.com | 2020-08-11 |  90000 |             1 |
|             1 | IT               |         109 | Rajesh     | Gupta     | rajesh.gupta@specialforce.com | 2020-08-11 |  90000 |             1 |
|             5 | Marketing        |         110 | Kavita     | Nair      | kavita.nair@specialforce.com  | 2021-02-07 |  52500 |             2 |
|             4 | Finance          |         110 | Kavita     | Nair      | kavita.nair@specialforce.com  | 2021-02-07 |  52500 |             2 |
|             3 | Sales            |         110 | Kavita     | Nair      | kavita.nair@specialforce.com  | 2021-02-07 |  52500 |             2 |
|             2 | HR               |         110 | Kavita     | Nair      | kavita.nair@specialforce.com  | 2021-02-07 |  52500 |             2 |
|             1 | IT               |         110 | Kavita     | Nair      | kavita.nair@specialforce.com  | 2021-02-07 |  52500 |             2 |
+---------------+------------------+-------------+------------+-----------+-------------------------------+------------+--------+---------------+
50 rows in set (0.00 sec)

mysql> select d.*, e.* from
    -> department d left join employee e;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near ''
 at line 2
mysql> select d.*, e.* from
    -> department d left join employee e
    -> ON d.department_id = e.department_id;
ERROR 1146 (42S02): Table 'sfpl.department_id' doesn't exist
mysql> SELECT d.*, e.*
    -> FROM department d
    -> LEFT JOIN employee e
    -> ON d.department_id = e.department_id;
+---------------+------------------+-------------+------------+-----------+-------------------------------+------------+--------+---------------+
| department_id | department_name  | employee_id | first_name | last_name | email                         | hire_date  | salary | department_id |
+---------------+------------------+-------------+------------+-----------+-------------------------------+------------+--------+---------------+
|             1 | IT               |         101 | Ravi       | Sharma    | ravi.sharma@specialforce.com  | 2017-05-15 |  55000 |             1 |
|             1 | IT               |         103 | Jyoti      | Verma     | jyoti.verma@specialforce.com  | 2020-11-02 |  60000 |             1 |
|             1 | IT               |         109 | Rajesh     | Gupta     | rajesh.gupta@specialforce.com | 2020-08-11 |  90000 |             1 |
|             2 | HR               |         102 | Neha       | Kapoor    | neha.kapoor@specialforce.com  | 2019-03-23 |  50400 |             2 |
|             2 | HR               |         107 | Jatin      | Reddy     | jatin.reddy@specialforce.com  | 2021-12-12 |  89250 |             2 |
|             2 | HR               |         110 | Kavita     | Nair      | kavita.nair@specialforce.com  | 2021-02-07 |  52500 |             2 |
|             3 | Sales            |         104 | Anil       | Patil     | anil.patil@specialforce.com   | 2018-09-18 |  70000 |             3 |
|             3 | Sales            |         106 | Sanjay     | Iyer      | sanjay.iyer@specialforce.com  | 2018-01-22 |  75000 |             3 |
|             4 | Finance          |         105 | Pooja      | Singh     | pooja.singh@specialforce.com  | 2021-06-10 |  40000 |             4 |
|             5 | Marketing        |         108 | Shreya     | Mehta     | shreya.mehta@specialforce.com | 2022-04-19 |  30000 |             5 |
+---------------+------------------+-------------+------------+-----------+-------------------------------+------------+--------+---------------+
10 rows in set (0.00 sec)

mysql>
```

Query 16



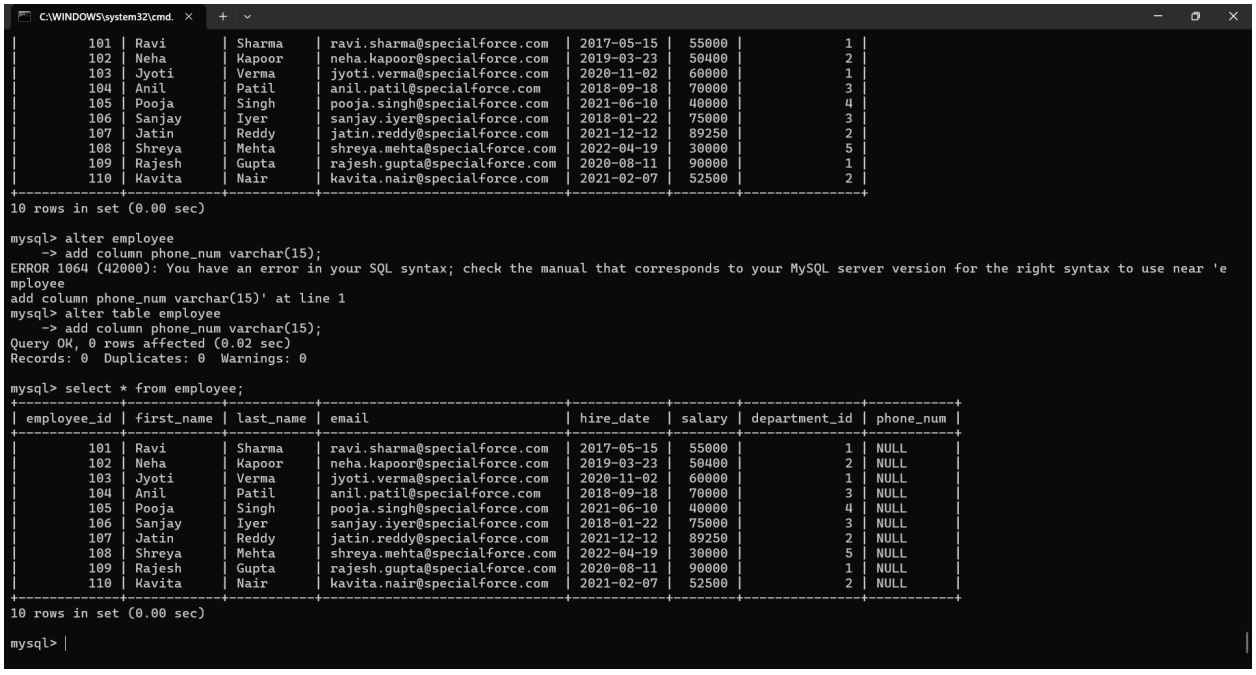
```
C:\WINDOWS\system32\cmd.

    -> FROM department d
    -> LEFT JOIN employee e
    -> ON d.department_id = e.department_id;
+---------------+-----------------+-------------+------------+-----------+-------------------------------+------------+--------+---------------+
| department_id | department_name | employee_id | first_name | last_name | email                         | hire_date  | salary | department_id |
+---------------+-----------------+-------------+------------+-----------+-------------------------------+------------+--------+---------------+
|             1 | IT              |         101 | Ravi       | Sharma    | ravi.sharma@specialforce.com  | 2017-05-15 |  55000 |             1 |
|             1 | IT              |         103 | Jyoti      | Verma     | jyoti.verma@specialforce.com  | 2020-11-02 |  60000 |             1 |
|             1 | IT              |         109 | Rajesh     | Gupta     | rajesh.gupta@specialforce.com | 2020-08-11 |  90000 |             1 |
|             2 | HR              |         102 | Neha       | Kapoor    | neha.kapoor@specialforce.com  | 2019-03-23 |  50400 |             2 |
|             2 | HR              |         107 | Jatin      | Reddy     | jatin.reddy@specialforce.com  | 2021-12-12 |  89250 |             2 |
|             2 | HR              |         110 | Kavita     | Nair      | kavita.nair@specialforce.com  | 2021-02-07 |  52500 |             2 |
|             3 | Sales           |         104 | Anil       | Patil     | anil.patil@specialforce.com   | 2018-09-18 |  70000 |             3 |
|             3 | Sales           |         106 | Sanjay     | Iyer      | sanjay.iyer@specialforce.com  | 2018-01-22 |  75000 |             3 |
|             4 | Finance         |         105 | Pooja      | Singh     | pooja.singh@specialforce.com  | 2021-06-10 |  40000 |             4 |
|             5 | Marketing       |         108 | Shreya     | Mehta     | shreya.mehta@specialforce.com | 2022-04-19 |  30000 |             5 |
+---------------+-----------------+-------------+------------+-----------+-------------------------------+------------+--------+---------------+
10 rows in set (0.00 sec)

mysql> select e.*, d.department_name, sum(e.salary)
    -> from employee e left join department d
    -> on e.department_id = d.department_id
    -> group by d.department_name ;
ERROR 1055 (42000): Expression #1 of SELECT list is not in GROUP BY clause and contains nonaggregated column 'sfpl.e.employee_id' which is not functionally
dependent on columns in GROUP BY clause; this is incompatible with sql_mode=only_full_group_by
mysql> select d.department_name, sum(e.salary)
    -> from employee e left join department d
    -> on e.department_id = d.department_id
    -> group by d.department_name ;
+-----------------+---------------+
| department_name | sum(e.salary) |
+-----------------+---------------+
| IT              |        205000 |
| HR              |        192150 |
| Sales           |        145000 |
| Finance         |         40000 |
| Marketing       |         30000 |
+-----------------+---------------+
5 rows in set (0.00 sec)

mysql>
```

Query 19



```
C:\WINDOWS\system32\cmd.

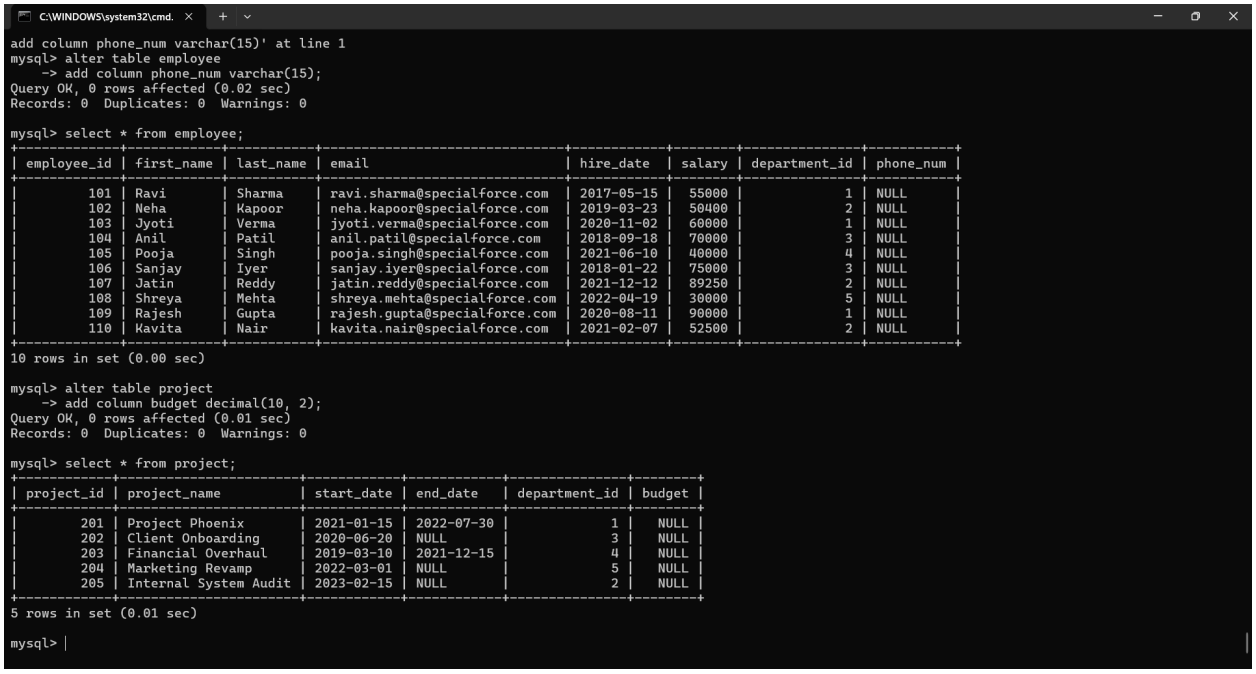
|         101 | Ravi       | Sharma    | ravi.sharma@specialforce.com  | 2017-05-15 |  55000 |             1 |
|         102 | Neha       | Kapoor    | neha.kapoor@specialforce.com  | 2019-03-23 |  50400 |             2 |
|         103 | Jyoti      | Verma     | jyoti.verma@specialforce.com  | 2020-11-02 |  60000 |             1 |
|         104 | Anil       | Patil     | anil.patil@specialforce.com   | 2018-09-18 |  70000 |             3 |
|         105 | Pooja      | Singh     | pooja.singh@specialforce.com  | 2021-06-10 |  40000 |             4 |
|         106 | Sanjay     | Iyer      | sanjay.iyer@specialforce.com  | 2018-01-22 |  75000 |             3 |
|         107 | Jatin      | Reddy     | jatin.reddy@specialforce.com  | 2021-12-12 |  89250 |             2 |
|         108 | Shreya     | Mehta     | shreya.mehta@specialforce.com | 2022-04-19 |  30000 |             5 |
|         109 | Rajesh     | Gupta     | rajesh.gupta@specialforce.com | 2020-08-11 |  90000 |             1 |
|         110 | Kavita     | Nair      | kavita.nair@specialforce.com  | 2021-02-07 |  52500 |             2 |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+
10 rows in set (0.00 sec)

mysql> alter employee
    -> add column phone_num varchar(15);
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'e
mployee
add column phone_num varchar(15)' at line 1
mysql> alter table employee
    -> add column phone_num varchar(15);
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> select * from employee;
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+-----------+
| employee_id | first_name | last_name | email                         | hire_date  | salary | department_id | phone_num |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+-----------+
|         101 | Ravi       | Sharma    | ravi.sharma@specialforce.com  | 2017-05-15 |  55000 |             1 | NULL      |
|         102 | Neha       | Kapoor    | neha.kapoor@specialforce.com  | 2019-03-23 |  50400 |             2 | NULL      |
|         103 | Jyoti      | Verma     | jyoti.verma@specialforce.com  | 2020-11-02 |  60000 |             1 | NULL      |
|         104 | Anil       | Patil     | anil.patil@specialforce.com   | 2018-09-18 |  70000 |             3 | NULL      |
|         105 | Pooja      | Singh     | pooja.singh@specialforce.com  | 2021-06-10 |  40000 |             4 | NULL      |
|         106 | Sanjay     | Iyer      | sanjay.iyer@specialforce.com  | 2018-01-22 |  75000 |             3 | NULL      |
|         107 | Jatin      | Reddy     | jatin.reddy@specialforce.com  | 2021-12-12 |  89250 |             2 | NULL      |
|         108 | Shreya     | Mehta     | shreya.mehta@specialforce.com | 2022-04-19 |  30000 |             5 | NULL      |
|         109 | Rajesh     | Gupta     | rajesh.gupta@specialforce.com | 2020-08-11 |  90000 |             1 | NULL      |
|         110 | Kavita     | Nair      | kavita.nair@specialforce.com  | 2021-02-07 |  52500 |             2 | NULL      |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+-----------+
10 rows in set (0.00 sec)

mysql>
```

Query 20



```
C:\WINDOWS\system32\cmd.
add column phone_num varchar(15)' at line 1
mysql> alter table employee
    -> add column phone_num varchar(15);
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

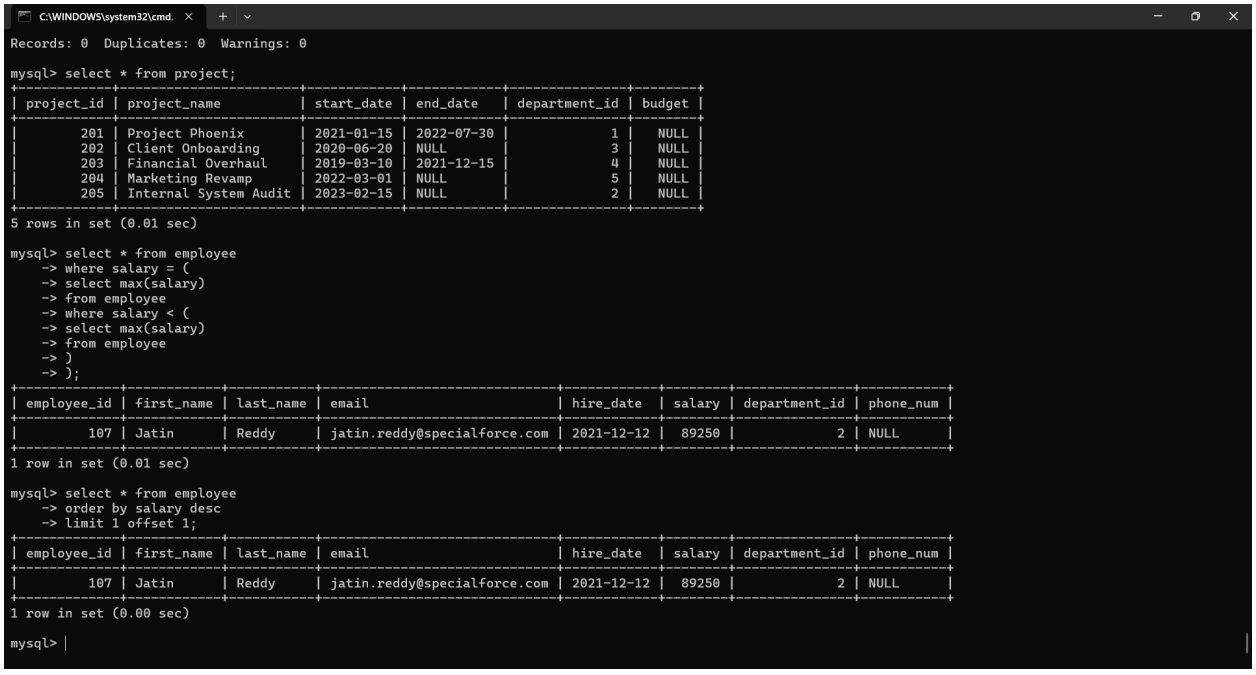
mysql> select * from employee;
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+------------+
| employee_id | first_name | last_name | email                         | hire_date  | salary | department_id | phone_num  |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+------------+
|         101 | Ravi       | Sharma    | ravi.sharma@specialforce.com  | 2017-05-15 |  55000 |             1 | NULL       |
|         102 | Neha       | Kapoor    | neha.kapoor@specialforce.com  | 2019-03-23 |  50400 |             2 | NULL       |
|         103 | Jyoti      | Verma     | jyoti.verma@specialforce.com  | 2020-11-02 |  60000 |             1 | NULL       |
|         104 | Anil       | Patil     | anil.patil@specialforce.com   | 2018-09-18 |  70000 |             3 | NULL       |
|         105 | Pooja      | Singh     | pooja.singh@specialforce.com  | 2021-06-10 |  40000 |             4 | NULL       |
|         106 | Sanjay     | Iyer      | sanjay.iyer@specialforce.com  | 2018-01-22 |  75000 |             3 | NULL       |
|         107 | Jatin      | Reddy     | jatin.reddy@specialforce.com  | 2021-12-12 |  89250 |             2 | NULL       |
|         108 | Shreya     | Mehta     | shreya.mehta@specialforce.com | 2022-04-19 |  30000 |             5 | NULL       |
|         109 | Rajesh     | Gupta     | rajesh.gupta@specialforce.com | 2020-08-11 |  90000 |             1 | NULL       |
|         110 | Kavita     | Nair      | kavita.nair@specialforce.com  | 2021-02-07 |  52500 |             2 | NULL       |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+------------+
10 rows in set (0.00 sec)

mysql> alter table project
    -> add column budget decimal(10, 2);
Query OK, 0 rows affected (0.01 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> select * from project;
+------------+----------------------+------------+------------+---------------+--------+
| project_id | project_name         | start_date | end_date   | department_id | budget |
+------------+----------------------+------------+------------+---------------+--------+
|        201 | Project Phoenix      | 2021-01-15 | 2022-07-30 |             1 |   NULL |
|        202 | Client Onboarding    | 2020-06-20 | NULL       |             3 |   NULL |
|        203 | Financial Overhaul   | 2019-03-10 | 2021-12-15 |             4 |   NULL |
|        204 | Marketing Revamp     | 2022-03-01 | NULL       |             5 |   NULL |
|        205 | Internal System Audit | 2023-02-15 | NULL       |             2 |   NULL |
+------------+----------------------+------------+------------+---------------+--------+
5 rows in set (0.01 sec)

mysql>
```

Query 21



```
C:\WINDOWS\system32\cmd.
Records: 0  Duplicates: 0  Warnings: 0

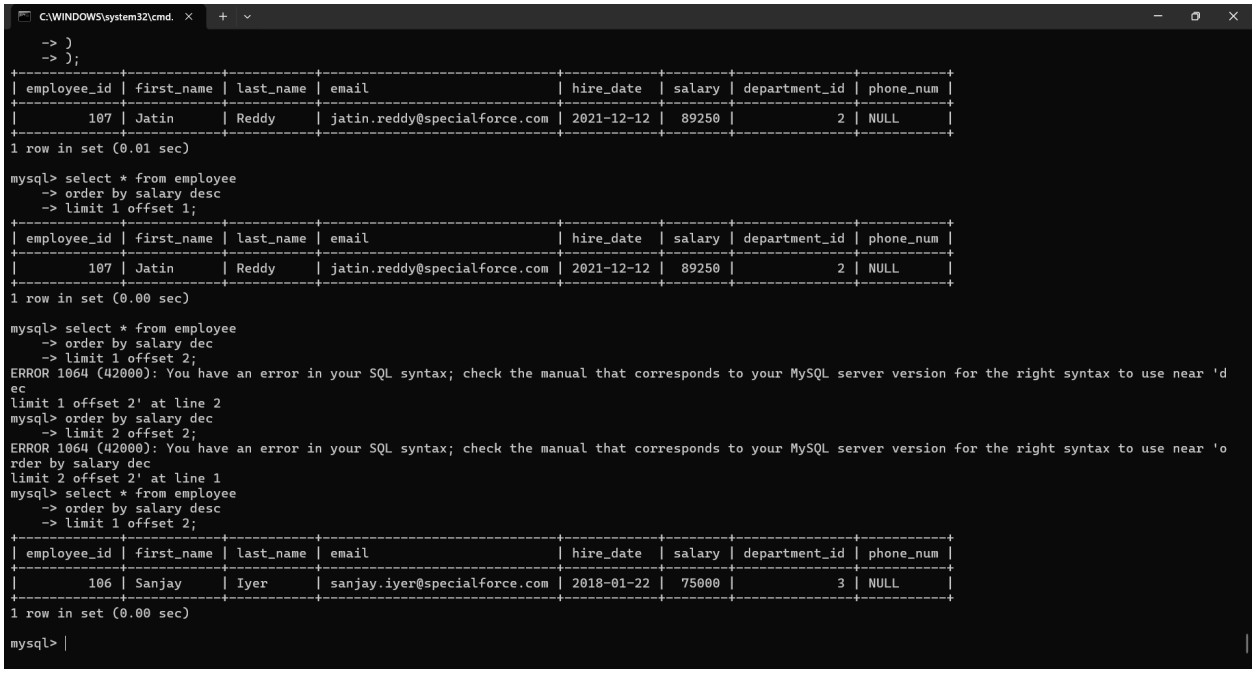
mysql> select * from project;
+------------+----------------------+------------+------------+---------------+--------+
| project_id | project_name         | start_date | end_date   | department_id | budget |
+------------+----------------------+------------+------------+---------------+--------+
|        201 | Project Phoenix      | 2021-01-15 | 2022-07-30 |             1 |   NULL |
|        202 | Client Onboarding    | 2020-06-20 | NULL       |             3 |   NULL |
|        203 | Financial Overhaul   | 2019-03-10 | 2021-12-15 |             4 |   NULL |
|        204 | Marketing Revamp     | 2022-03-01 | NULL       |             5 |   NULL |
|        205 | Internal System Audit | 2023-02-15 | NULL       |             2 |   NULL |
+------------+----------------------+------------+------------+---------------+--------+
5 rows in set (0.01 sec)

mysql> select * from employee
    -> where salary = (
    -> select max(salary)
    -> from employee
    -> where salary < (
    -> select max(salary)
    -> from employee
    -> )
    -> );
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+------------+
| employee_id | first_name | last_name | email                         | hire_date  | salary | department_id | phone_num  |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+------------+
|         107 | Jatin      | Reddy     | jatin.reddy@specialforce.com  | 2021-12-12 |  89250 |             2 | NULL       |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+------------+
1 row in set (0.01 sec)

mysql> select * from employee
    -> order by salary desc
    -> limit 1 offset 1;
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+------------+
| employee_id | first_name | last_name | email                         | hire_date  | salary | department_id | phone_num  |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+------------+
|         107 | Jatin      | Reddy     | jatin.reddy@specialforce.com  | 2021-12-12 |  89250 |             2 | NULL       |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+------------+
1 row in set (0.00 sec)

mysql>
```

Query 22



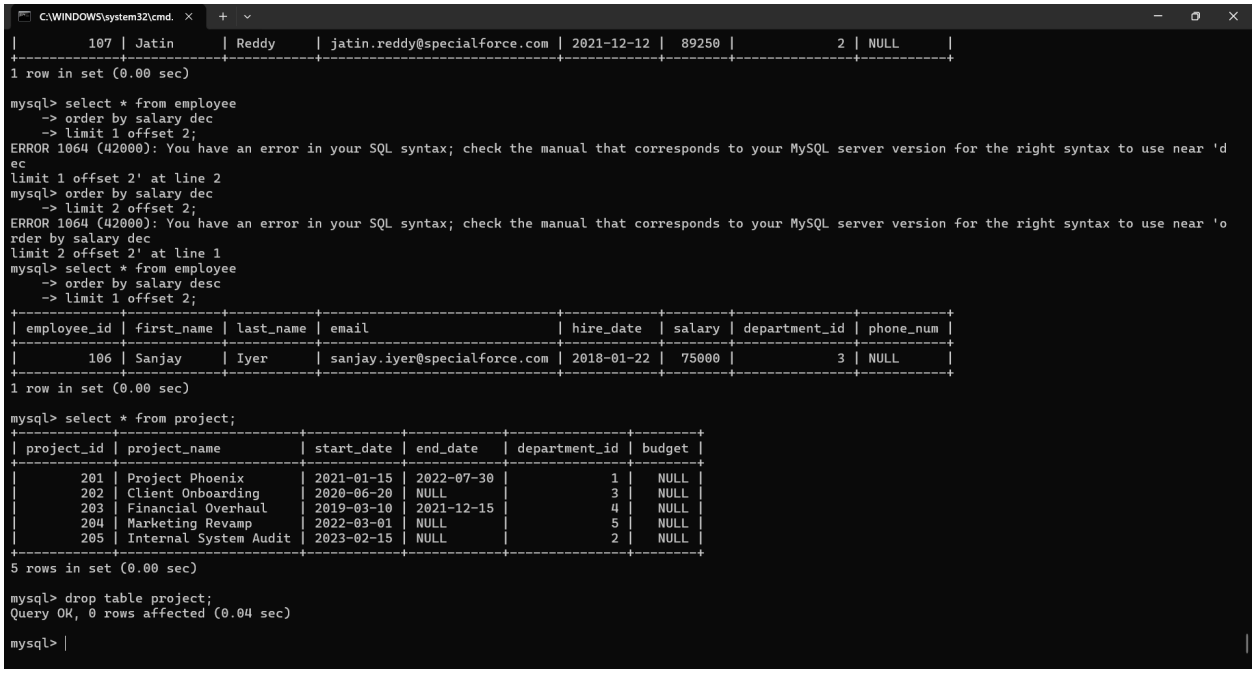
```
C:\WINDOWS\system32\cmd.
    -> )
    -> );
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+-----------+
| employee_id | first_name | last_name | email                         | hire_date  | salary | department_id | phone_num |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+-----------+
|         107 | Jatin      | Reddy     | jatin.reddy@specialforce.com | 2021-12-12 |  89250 |             2 | NULL      |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+-----------+
1 row in set (0.01 sec)

mysql> select * from employee
    -> order by salary desc
    -> limit 1 offset 1;
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+-----------+
| employee_id | first_name | last_name | email                         | hire_date  | salary | department_id | phone_num |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+-----------+
|         107 | Jatin      | Reddy     | jatin.reddy@specialforce.com | 2021-12-12 |  89250 |             2 | NULL      |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+-----------+
1 row in set (0.00 sec)

mysql> select * from employee
    -> order by salary dec
    -> limit 1 offset 2;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'd
ec
limit 1 offset 2' at line 2
mysql> order by salary dec
    -> limit 2 offset 2;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'o
rder by salary dec
limit 2 offset 2' at line 1
mysql> select * from employee
    -> order by salary desc
    -> limit 1 offset 2;
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+-----------+
| employee_id | first_name | last_name | email                         | hire_date  | salary | department_id | phone_num |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+-----------+
|         106 | Sanjay     | Iyer      | sanjay.iyer@specialforce.com | 2018-01-22 |  75000 |             3 | NULL      |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+-----------+
1 row in set (0.00 sec)

mysql>
```

Query 23/18



```
C:\WINDOWS\system32\cmd.
|         107 | Jatin      | Reddy     | jatin.reddy@specialforce.com | 2021-12-12 |  89250 |             2 | NULL      |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+-----------+
1 row in set (0.00 sec)

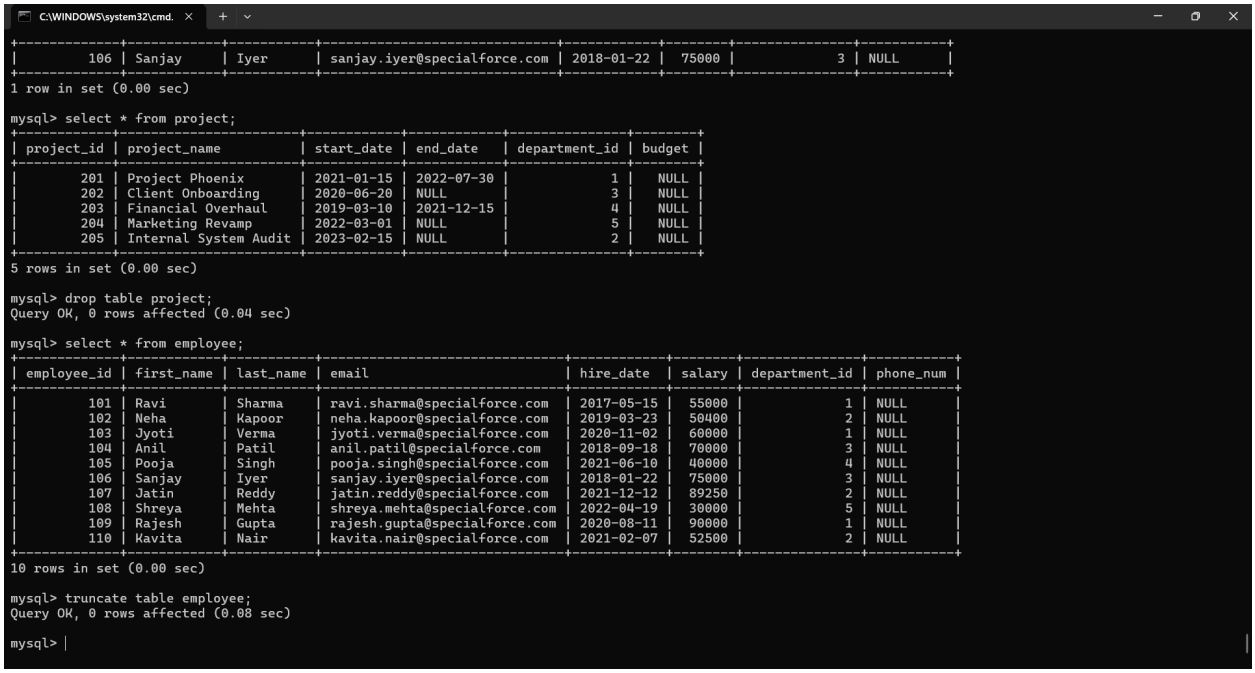
mysql> select * from employee
    -> order by salary dec
    -> limit 1 offset 2;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'd
ec
limit 1 offset 2' at line 2
mysql> order by salary dec
    -> limit 2 offset 2;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'o
rder by salary dec
limit 2 offset 2' at line 1
mysql> select * from employee
    -> order by salary desc
    -> limit 1 offset 2;
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+-----------+
| employee_id | first_name | last_name | email                         | hire_date  | salary | department_id | phone_num |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+-----------+
|         106 | Sanjay     | Iyer      | sanjay.iyer@specialforce.com | 2018-01-22 |  75000 |             3 | NULL      |
+-------------+------------+-----------+-------------------------------+------------+--------+---------------+-----------+
1 row in set (0.00 sec)

mysql> select * from project;
+------------+-----------------------+------------+------------+---------------+--------+
| project_id | project_name          | start_date | end_date   | department_id | budget |
+------------+-----------------------+------------+------------+---------------+--------+
|        201 | Project Phoenix       | 2021-01-15 | 2022-07-30 |             1 |   NULL |
|        202 | Client Onboarding     | 2020-06-20 | NULL       |             3 |   NULL |
|        203 | Financial Overhaul    | 2019-03-10 | 2021-12-15 |             4 |   NULL |
|        204 | Marketing Revamp      | 2022-03-01 | NULL       |             5 |   NULL |
|        205 | Internal System Audit | 2023-02-15 | NULL       |             2 |   NULL |
+------------+-----------------------+------------+------------+---------------+--------+
5 rows in set (0.00 sec)

mysql> drop table project;
Query OK, 0 rows affected (0.04 sec)

mysql>
```

Query 24/17



```
C:\WINDOWS\system32\cmd.

+-------------+-------------+-----------+-------------------------------+------------+--------+---------------+-----------+
|         106 | Sanjay      | Iyer      | sanjay.iyer@specialforce.com | 2018-01-22 |  75000 |             3 | NULL      |
+-------------+-------------+-----------+-------------------------------+------------+--------+---------------+-----------+
1 row in set (0.00 sec)

mysql> select * from project;
+------------+-----------------------+------------+------------+---------------+--------+
| project_id | project_name          | start_date | end_date   | department_id | budget |
+------------+-----------------------+------------+------------+---------------+--------+
|        201 | Project Phoenix       | 2021-01-15 | 2022-07-30 |             1 |   NULL |
|        202 | Client Onboarding     | 2020-06-20 | NULL       |             3 |   NULL |
|        203 | Financial Overhaul    | 2019-03-10 | 2021-12-15 |             4 |   NULL |
|        204 | Marketing Revamp      | 2022-03-01 | NULL       |             5 |   NULL |
|        205 | Internal System Audit | 2023-02-15 | NULL       |             2 |   NULL |
+------------+-----------------------+------------+------------+---------------+--------+
5 rows in set (0.00 sec)

mysql> drop table project;
Query OK, 0 rows affected (0.04 sec)

mysql> select * from employee;
+-------------+------------+-----------+--------------------------------+------------+--------+---------------+-----------+
| employee_id | first_name | last_name | email                          | hire_date  | salary | department_id | phone_num |
+-------------+------------+-----------+--------------------------------+------------+--------+---------------+-----------+
|         101 | Ravi       | Sharma    | ravi.sharma@specialforce.com   | 2017-05-15 |  55000 |             1 | NULL      |
|         102 | Neha       | Kapoor    | neha.kapoor@specialforce.com   | 2019-03-23 |  50400 |             2 | NULL      |
|         103 | Jyoti      | Verma     | jyoti.verma@specialforce.com   | 2020-11-02 |  60000 |             1 | NULL      |
|         104 | Anil       | Patil     | anil.patil@specialforce.com    | 2018-09-18 |  70000 |             3 | NULL      |
|         105 | Pooja      | Singh     | pooja.singh@specialforce.com   | 2021-06-10 |  40000 |             4 | NULL      |
|         106 | Sanjay     | Iyer      | sanjay.iyer@specialforce.com   | 2018-01-22 |  75000 |             3 | NULL      |
|         107 | Jatin      | Reddy     | jatin.reddy@specialforce.com   | 2021-12-12 |  89250 |             2 | NULL      |
|         108 | Shreya     | Mehta     | shreya.mehta@specialforce.com  | 2022-04-19 |  30000 |             5 | NULL      |
|         109 | Rajesh     | Gupta     | rajesh.gupta@specialforce.com  | 2020-08-11 |  90000 |             1 | NULL      |
|         110 | Kavita     | Nair      | kavita.nair@specialforce.com   | 2021-02-07 |  52500 |             2 | NULL      |
+-------------+------------+-----------+--------------------------------+------------+--------+---------------+-----------+
10 rows in set (0.00 sec)

mysql> truncate table employee;
Query OK, 0 rows affected (0.08 sec)

mysql>
```